

- Chakrabarty, Ananda, 1:61, 4:129
- Chaotropic salts, 3:221
- Chaperones, molecular (chaperonins), 1:116–119
discovery of, 1:116
energy requirements, 1:116–117, 1:117
functions, 1:116–117, 2:164, 3:203, 4:168
Hsp70 and Hsp60 systems, 1:116, 1:117, 1:118
and human disease, 1:118
- Charcot-Marie-Tooth disease, 4:4
- Chargaff, Erwin, 1:250–251, 4:172
- Chargaff's ratios, 1:250–251, 4:172
- Charge density, defined, 2:46
- Chase, Alfred, 1:204
- Chase, Martha, 3:104, 3:105, 4:120
- Cheapdate mutant fruit flies, 1:5–6
- Cheetahs, population bottlenecks, 1:189
- Chelation
bioremediation role, 1:60
DNA purification role, 3:221
- Chemicals, toxic. *See* Industrial toxins; Mutagens
- Chemiosmotic gradients, 3:51
- Chemistry, combinatorial. *See* Combinatorial chemistry
- Chemokine receptors
of retroviruses, 4:36
structure, 1:55
- Chemolithotrophs, described, 2:13
- Chemotaxis, defined, 2:158
- Chemotherapy
and immunodeficiency, 4:74
pharmacogenomics role, 3:146
- Chi sequences, *E. coli*, 2:117
- Chiasmata, in crossing over, 1:194, 1:195, 3:26
- Chickenpox
AIDS and, 2:155
characteristics, 4:165
- Children's Bureau (Maternal and Child Health Bureau), PKU screening, 3:219
- Chimeras, embryonic stem cells and, 2:3–5, 2:5, 4:125
- Chimney sweeps, and cancer, 1:98–99
- Chimpanzee foamy spumavirus, 4:35
- Chimpanzees, genetic diversity, 3:167–168, 3:168
- Chiron Corporation, 1:72
- Chi-square test, 3:194–195
- Chitin, in cell walls, 1:36
- Chlamydia trachomatis*, genome characteristics, 1:142
- Chlamydia pneumoniae*, genome characteristics, 1:142
- Chlamydomonas*, as model organism, 3:61
- Chlorophyll
albino plants, 3:10
function, 1:112
- Chloroplasts
defined, 2:194, 3:149, 4:42
ribosomes, 4:42
ribozymes, 4:44
structure and function, 1:112, 2:194, 4:50
transplastomics, 2:108
- Chloroplasts, genome
base pairs, number, 2:116
cpDNA, 2:194, 2:195, 2:196, 2:198, 3:55
distinguished from eukaryotes, 2:112, 2:195–196
evolutionary origin, 2:12, 2:108, 2:116, 2:194–196, 3:56–57
as evolutionary study tool, 2:198
inheritance patterns, 2:196–197
vegetative segregation, 2:197
- Cholera, genetic studies, 1:139–140
- Cholesterol
diabetes and, 1:212
familial hypercholesterolemia and, 2:57
hypercholesterolemia, 4:4–5
metabolism, as quantitative trait, 4:2
- Chondrocytes
achondroplasia and, 2:130
defined, 2:130
- Chorionic villus sampling (CVS)
defined, 1:258, 2:97
for Down syndrome, 1:258
uses, 3:184, 3:187
- Chorismate, as tryptophan metabolite, 3:132
- Chromatids
defined, 1:126, 3:106
See also Chromosomes; Crossing over (recombination)
- Chromatin
ATP-dependent complexes, 1:138–139
condensation, 2:63, 2:65, 2:112–113, 2:164, 3:25
defined, 1:126, 2:63
disruption of, 1:137–138
heterochromatin-euchromatin conversion, 1:134–135, 2:65
methylation of, 3:77
in nucleus, 3:120, 3:120, 3:121
structure, 1:133–134
- Chromatin-remodeling factors (enzymes), 1:137–138, 4:111
- Chromatography
in mass spectrometry analysis, 3:19
as protein sequencing tool, 3:197
- Chromatosomes, defined, 1:133
- Chromomeres, 1:136
- Chromophores, role in color vision, 1:171
- Chromosomal aberrations, 1:119–125
assays for, 3:92–93, 3:182–185
birth defects, 1:75, 1:79, 3:182
breakage-fusion-bridge cycle, 3:21, 3:22
candidate gene studies of, 2:59
causes, 1:119, 1:123
centromeres lacking, 3:97
consequences, 4:67–68
deletions, 1:122, 1:123, 1:124
DNA methylation role, 3:49
duplications, 1:122, 1:123–124, 1:124, 3:97
gene evolution role, 2:28
history of discoveries, 1:119
intersex organisms, 1:25, 3:77, 3:164, 4:80, 4:82, 4:127, 4:175
inversions, 1:122, 1:124–125, 1:124, 3:97
nomenclature, 3:107
polyspermy, 2:35, 3:163–164
prenatal diagnosis, 2:97, 3:182–187
recurrence risk, 1:125
techniques for studying, 1:119
translocations, 1:122–123, 1:124, 3:96–97, 3:127, 3:128–129, 4:154
uniparental disomy, 2:185
of Y chromosomes, 4:177
See also Aneuploidy; Nondisjunction; Polyploidy
- Chromosomal aberrations, specific disorders
Angelman's syndrome, 1:123, 2:185, 2:205
cancer, 3:127, 3:130
Klinefelter's syndrome, 1:2, 1:79, 1:121–122, 3:111, 4:80, 4:174
Prader-Willi syndrome, 1:123, 2:185, 2:205
See also Down syndrome; Fragile X syndrome; Turner's syndrome
- Chromosomal banding, 1:125–129, 3:106, 3:107
bands, defined, 1:119
C-banding, 1:126, 1:128

- cell synchronization, 1:128
DAPI dyes, 1:127
FISH technique, 1:128–129, 1:145
G-banding, 1:127, 1:127, 1:128
GTG banding, 1:127
NOR-staining, 1:128
numbering, 1:132, 1:135–136
polytene chromosomes, 1:136–137, 2:44
Q-banding, 1:119, 1:127, 1:135
R-banding, 1:128
Chromosomal territories, 3:121–122, 3:122
Chromosomal theory of inheritance, history, 1:129–132, 2:42–43, 2:50, 3:102
See also DNA structure and function, history; Gene nature, history; McClintock, Barbara; Mendelian genetics; Morgan, Thomas Hunt; Muller, Hermann
Chromosome mapping. *See* Mapping genes
Chromosome painting (spectral karyotyping), 3:183
Chromosome terminology, 3:106–107
Chromosomes, artificial, 1:144–146
 bacterial (BACs), 1:144–146, 1:222–224
 mammalian (MACs), 1:145, 1:146
 yeast (YACs), 1:144–145, 1:222–224
Chromosomes, circular
 Archaea, 1:36
 conjugation instructions, 1:182
 eubacteria, 1:140, 2:14–15, 3:132, 4:12
 mitochondrial and chloroplast genomes, 2:116, 2:196, 3:55–56
 prokaryotes, 1:139–140, 1:143, 2:14–15, 2:116, 4:12
 telomeres lacking, 4:104
 See also Plasmids
Chromosomes, eukaryotic, 1:105, 1:132–139, 1:133
 acentric fragments, 1:114
 acrocentric, 1:125
 autosomes, 1:3, 1:79, 1:119, 1:196–197
 chromatin-remodeling complexes, 1:138–139
 dicentric, 1:114
 distinguished from prokaryotes, 1:132, 1:139
 function, 1:144
 haploid complements, 2:113, 3:24
 heterochromatin-euchromatin conversion, 1:134–135, 2:65
 higher-order organization, 1:132–134, 1:134
 holocentric (holokinetic), 1:114
 karyotypes, 1:127, 3:108
 “knobs,” staining, 3:10
 in nucleus, 1:112, 1:132–133, 3:119–126
 number range, 2:113–114
 p and q arms, 1:135–136, 2:39, 3:106–107
 polyspermy and, 2:35
 polytene, 1:126, 1:136–137, 1:136, 2:43–44
 puffing, 1:126, 1:136–137, 1:136
 ribonuclease role, 3:114–115
 structure, 1:125–126, 1:132, 1:134, 1:135–136, 1:249
 See also Autosomes; Cells, eukaryotic; Chromosomes, sex; Crossing over (recombination); Eukaryotes; Meiosis; Mitosis; Polyploidy; Telomeres
Chromosomes, homologous
 defined, 1:123
 See also Crossing over (recombination)
Chromosomes, origin of term, 1:130, 1:132
Chromosomes, prokaryotic, 1:139–144
 distinguished from eukaryotes, 1:132, 1:139
 domains, 1:140–141
 function, 1:144
 minimal-gene-set concept, 1:143
 number, range, 2:113
 operons, 1:142
 replication, 1:141, 2:15
 shape, 1:139–140, 1:143, 2:14–15, 2:116
 supercoiled DNA, 1:141, 1:143
 transposable genetic elements, 4:144–145
 See also Cells, prokaryotic; Conjugation; Mitosis; Plasmids; Prokaryotes
Chromosomes, sex, 2:114
 aneuploidy, 1:121, 3:109–112, 4:80
 as barrier to polyploidy, 3:164
 evolution of, 4:174, 4:177
 human, number, 1:79, 1:119, 1:137, 1:196, 2:33, 2:113, 2:174, 2:199, 3:24, 3:76, 3:106, 3:108–109, 3:111, 3:182
 intersex organisms, 1:25, 3:77, 3:164, 4:80, 4:82, 4:127, 4:175
 lacking in plants, 3:164
 monosomy, 3:110–111
 mutations, 1:120
 See also Sex determination; Trisomy; X chromosomes; Y chromosomes
Chronic granulomatous disease, genomic screening, 2:168
Chrysanthemums, polyploid, 3:166
Cilia, 1:105
Ciliates, ploidy in, 2:115
Cip/Kip family proteins, 1:107, 1:108
Ciproflaxin (Cipro), 2:15
Circadian, defined, 1:36
Cis regulatory elements, 4:114
Citric acid, as phytochelator, 1:60
CJD (Creutzfeldt-Jakob disease), 3:188, 3:189–190
C-*jun* transcription factor, 3:130
CKIs (CDK inhibitors), 1:106–107
Classical hybrid genetics, 1:146–149
 allele inheritance patterns, 1:147–148
 predictive value, 1:146–147, 1:148–149
 Punnett squares, 1:148
 X-linked inheritance studies, 1:130–131
 See also Chromosomal theory of inheritance, history; Gene nature, history; Mendelian genetics
Classification analysis, DNA microarray tools, 1:228
CIB chromosomes, 3:91–92
Cleave/cleavage, defined, 1:220, 3:113
Cleft chin, 2:202
Cleft lip/palate (CL/P), 1:75, 1:77, 1:78, 2:130
CLIA (Clinical Laboratory Improvement Act), 3:218
Climatic bottlenecks, 3:168–169, 3:169
Clinical geneticists, 1:149–151
 to diagnose birth defects, 1:74
 as genetic counselors, 2:90, 2:91–92
 genetic disease research, 1:213–215
 vs. medical geneticist, 1:149
 probability tools, 3:193–194
Clinical Laboratory Improvement Act (CLIA), 3:218
Clinical trials
 databases, 1:102–103
 defined, 1:69, 2:82
 gene therapy, ethical issues, 1:69, 2:81, 2:83

- Clinical trials (continued)
 genomics industry role, 2:124
 informed consent, 1:69, 2:81–82
- Clomiphene citrate, 4:19
- Cloning, defined, 1:152, 1:158
- Cloning, ethical issues, **1:158–161**
 bioethical concerns, 1:159, 1:160
 cloning bans and moratoriums, 1:71, 1:158–160, 2:3
 context, 1:158
 eugenics movement, 2:21
 extinct animal species, 4:25
 human cloning, 1:69, 1:158–160, 1:161, 2:3, 2:6
 misconceptions, 1:160
- Cloning genes, **1:152–158**
 CFTR gene, 2:60
 vs. cloning organisms, 1:152, 1:161
 DNA library tools, 1:222–224
 DNA purification tools, 3:220–221
 E. coli as tool, 2:10
 electrophoresis tools, 2:50
 expression cloning, 1:157, 3:152
 genomic *vs.* cDNA clones, 1:153–154
 Human Genome Project role, 2:171–172
 as identification tool, 1:154–155
 medical and industrial importance, 1:153–154
 as oncogene study tools, 3:127
 PCR as tool, 3:158
 plasmid vectors, 3:152
 positional cloning, 1:155–157, 1:202
 purposes, 1:152
 reverse transcriptase and, 1:153, 1:154, 4:40, 4:128
 techniques, 1:152–153, 4:128
 transposons as tools, 4:148
 triplet repeats, 4:152
 Werner's syndrome gene, 1:3
- Cloning organisms, **1:161–165**
 amphibians, 1:158, 1:161–163
 asexual reproduction, 1:161
 vs. cloning genes, 1:152, 1:161
 embryonic stem cell research, 2:3–6
 endangered species, 1:158, 1:161, 1:164
 history, 1:158, 1:161
 livestock, 1:160, 1:162, 1:163–164
 mice, 1:160, 4:61
 pets, 1:158–159, 1:159
 pigs, for organ transplants, 4:143
 plants, 1:161, 3:166
 reproductive cloning, 1:159–160
 technical problems, 1:160, 1:164–165, 2:3
 therapeutic cloning, 1:159–160, 1:161, 1:163–164
- Cloning vectors, 1:70–71, 1:152–153, 2:4, 4:128
- Clostridium*
 characteristics, 2:12, 2:13
 endospores, 2:12, 2:14
- Clotting factors. *See* Blood clotting factors
- Club foot, 1:75, 1:77, 1:80
- Clustering analysis, 1:227
- CML (chronic myelogenous leukemia), 3:127, 3:129, 4:154
- C-myc* gene, mutations, 3:96, 4:146
- CNS disease, causal organism, 4:165
- Coagulation factor IX, 1:163–164
- Cob* genes, 2:194
- Cocci bacterial form, 2:13, 2:14
- Cockayne's syndrome, progeroid aspects, 1:2
- CODIS (Combined DNA Indexing System), 1:235, 1:236, 1:237
- Codominance
 AB blood types, 3:36
 defined, 2:200, 3:36
- Codons
 anticodons, 2:84, 4:50, 4:137–138
 defined, 2:84, 2:157, 3:96, 3:200, 3:209, 4:4
 genetic code, table, 2:85, 4:137
 possible combinations, 2:84, 4:135
 sequence comparisons, mammals, 2:157
 start (promoter), 2:52, 2:52, 2:84, 3:135, 4:4, 4:36, 4:109, 4:137, 4:138
 stop (terminator), 1:13, 2:84, 2:85, 2:87, 3:96, 3:209, 4:110, 4:137, 4:138
 synonyms, 2:84, 2:85
 See also Genetic code
- Cofactors, metabolic role, 3:37, 3:39, 3:39, 3:43
- Cohen, Stanley, 1:71, 1:152
- Coiled-coil motif, 3:201
- Col plasmids, 3:151
- Colchicine, 3:165, 3:166
- Cold Spring Harbor Laboratory, 1:204, 4:173
- Colectomy, defined, 1:168
- Colitis, and colon cancer, 1:170
- Collagen proteins
 cartilage matrix protein mutations and, 2:131–132
 defined, 2:29
 function, 2:201
- Collagen proteins, function, 3:199
- Collagen-structure disorders, 2:201
- College professors, **1:165–166**, 1:166, 2:1–2
- Collins, Francis, 2:174
- Colon cancer. *See* Cancer, colon
- Colon crypts
 defined, 1:103
 replication, 1:103
- Colonization bottlenecks. *See* Founder effect
- Colonoscopies, 1:170
- Colony-stimulating factors, cost and access concerns, 1:69
- Color vision and color blindness. *See* Vision, color
- Colorado Adoption Study (CAP), 2:210
- Combinatorial chemistry, **1:173–177**
 approach, 1:173–174
 drug development role, 2:124–125
 drug targets, 1:175
 genetics applications, 1:174
 high-throughput screening tools, 1:175, 1:176, 2:149–150
 SELEX, 1:176–177
- Combinatorial control of development, 1:207
- Combinatorial gene regulation, 2:65, 2:66
- Combined DNA Indexing System (CODIS), 1:235, 1:236, 1:237
- Common cold, causal organism, 4:165
- COMP (cartilage oligomeric matrix protein), 2:131–132
- Competence factors, 4:121
- Complementary
 antisense nucleotides as, 1:29, 2:106
 base pairs as, 1:198, 1:217, 1:225, 1:230, 2:24, 2:50, 2:186, 3:94, 3:118–119, 3:155, 4:107
 defined, 1:29, 2:186, 3:118, 4:53
 molecular probes, 1:88–89
- Complementary DNA. *See* DNA, complementary (cDNA)
- Complex (polygenic) traits, **1:177–181**
 defined, 1:178, 2:206
 disease role, 1:178–179
 distinguished from simple traits, 1:177, 2:60–61
 DNA microarray analysis, 1:228
 emergence, 2:9
 epistatic interactions, 2:9

- eugenics and, 2:20
 genomics tools to study, 2:122
 inheritance patterns, 1:77–78, 1:213–214, 2:206
 in Mendelian genetics, 3:102
 modifier genes, 1:131
 multifactorial, 1:178, 2:206
 oligogenic, 1:178
 in plant breeding, 2:106
See also Gene-environment interactions; Quantitative (continuous) traits
- Complex (polygenic) traits, specific
 addiction, 1:4–6, 1:77
 aging, 1:6–8
 Alzheimer's disease, 1:15–18, 1:178, 1:180–181, 1:213, 2:61
 behavior, 1:48, 1:51
 cancer, 1:213
 cardiovascular disease, 1:101–103, 1:178, 1:213, 2:102
 diabetes, 1:178, 1:210, 1:213, 2:102
 eye color, 2:33
 intelligence, 2:206, 2:211, 3:81
 longevity, 3:81
 psychiatric disorders, 1:213, 3:214–216
 sexual orientation, 4:83–84
- Computational biologists, 1:54–56, 1:181–182, 3:196
- Computer technology
 automated sequencers, 1:44, 2:173–174
 bioinformatics tools, 1:52–56, 1:53
 BLAST, 2:156, 2:157, 2:212, 3:196
 computational biologists and, 1:181–182
 genetic microchips, 2:119–120
 homologous sequence analysis, 2:157
 information systems managers, 2:192–194
 Internet, 2:211–212
 in linkage and recombination studies, 3:7
 in mass spectrometry analysis, 3:18
 as molecular anthropology tool, 3:67
 pedigree drawing and recording, 3:140
 protein folding predictions, 2:177
 as proteomics tool, 3:208–209
 RNA structure analysis, 4:48
 statistical geneticists and, 4:94
- COMT* (Catechol-O-Methyltransferase) gene, 1:41
- Concatamers, function, 4:118
- Concentration gradients, role in development, 1:206
- Cones, role in color vision, 1:170–171
- Confidentiality. *See* Privacy
- Conformation
 defined, 3:201
 of proteins, 3:200–202, 3:204
- Congenital adrenal hyperplasia, newborn screening, 3:176, 3:177
- Congenital generalized lipodystrophic diabetes (CGL), 1:211, 2:203
- Conifers, inheritance of mitochondrial and chloroplast genes, 2:197
- Conjugation, 1:182–185
 and antibiotic resistance, 1:26–28, 1:140, 1:185
 defined, 1:27, 2:15, 4:123
 DNase and, 4:123
E. coli, 1:141–142, 1:183, 1:183, 1:185
 and escaped transgenes, 1:68
 eubacteria, 1:141–142
 plasmid role, 1:140–142, 1:182–185, 1:184, 2:117, 3:151
 sex pilus, 1:183, 1:183, 1:184, 2:15, 3:151
- Connective tissues, sarcomas, 1:93
- Connexin 26* gene, 1:76
- Consanguinity. *See* Inbreeding
- Conservation biology, genetic approaches, 1:186–190
 applications, 1:186–187, 1:190–191
 cloning extinct species, 4:25
 Hardy-Weinberg equilibrium, 2:135–136
 illegal trading concerns, 1:186, 1:189–190
 threatened population categories, 1:186
 tools, 1:187–188, 1:190–191
 units of conservation, 1:189, 1:190
See also Genetic drift; Natural selection; Population bottlenecks; Speciation
- Conservation geneticists, 1:190–192
- Contigs, gene discovery role, 2:60
- Continuous traits. *See* Quantitative traits
- Convention on International Trade in Endangered Species of Wild Fauna and Flora, 1:190
- Cook, Captain James, 1:63
- Cooley's anemia (thalassemia major), 2:140
- Cooper, Geoff, 3:127
- Corn. *See* Maize
- Corn borers, *Bacillus thuringiensis* and, 1:58, 1:64
- Coronary artery disease
 early-onset, 1:101–103, 1:102
 gene therapy, 2:74
See also Cardiovascular disease
- Correns, Carl, 3:102, 3:103
- Corynebacterium*, to synthesize vitamin C, 1:63
- Cosmetic testing, Ames test, 1:20–21
- Cotton, genetically engineered, 1:9, 2:107, 2:108, 4:134
- Cows. *See* Livestock
- CpG dinucleotides, methylation of, 3:46–49, 3:47
- CpG islands, 3:47, 3:48
- Cre recombinase enzyme, 2:73
- Creatine kinase, isoforms, 2:53
- CREB (cyclic AMP response element binding protein), signal transduction role, 4:88, 4:89
- Creighton, Harriet, 3:10, 3:21
- Creutzfeldt-Jakob disease (CJD), 3:188, 3:189–190
- Cri du Chat* syndrome, 1:123
- Crick, Francis, 1:192–194
 DNA structure research, 1:62, 1:193, 1:251, 2:50, 2:120, 2:171, 3:105, 4:172
 education and training, 1:192
 embryology and neurobiology studies, 1:193–194
 on flow of genetic information, 4:39
 on genetic coding, 1:252
 wobble hypothesis, 2:85, 4:4
- Criminal investigations. *See* Forensics
- Cristae, in mitochondrial membranes, 3:55, 3:56
- Critically Endangered population status, 1:186
- Crocodiles, illegal trade concerns, 1:189
- Crohn's disease, and colon cancer, 1:170
- Crossing over (recombination), 1:194–197
 chiasmata role, 1:194, 1:195, 3:26
CIB chromosome to prevent, 3:91–92
 defined, 3:4–5
 discovery, 1:131, 3:10, 3:21, 3:75, 3:105
 DNA comparison studies, 3:65

- Crossing over (recombination) (continued)
 gene amplification, 2:28, 2:28
 as gene mapping tool, 1:197, 2:59, 3:11–13, 3:26, 3:83
 genetic diversity role, 1:195–196, 3:24, 3:25, 3:26, 3:29, 3:60
 homologues and chromatids, 1:194, 1:195
 interference, 1:197, 3:12–13
 knock-out and knock-in genes, 3:91
 lacking in Y chromosomes, 3:65, 4:174, 4:177, 4:178
 mechanisms, 1:194–195, 1:195, 1:196, 2:28, 3:24, 3:25, 3:26, 3:29
 pseudogenes, 3:210, 3:210
 recombination fraction, 3:6–7, 3:12–13
 recombination rates, 1:196, 1:197, 3:26
 in transposons, 4:148
 unequal, 1:197, 3:7
 X-Y crossovers, 1:196–197
See also Linkage and recombination (linkage analysis); Meiosis
- Cross-matching blood, 1:83
- Crown gall disease (*Agrobacterium tumefaciens*), 1:9, 1:35, 4:132
- CRS (congenital rubella syndrome). *See* Rubella
- Cryogenic electron microscopy (cryoEM), 4:165
- Cryopreservation
 defined, 4:20
 in reproductive technology, 4:26
- C-src* (cell-sarcoma) gene, 1:100
- C-terminus, of amino acids/proteins, 3:181, 3:197–198, 3:207
- Cultivars, defined, 1:146
- Cumulina the cloned mouse, 1:160
- Cumulus oophorus, 2:35
- CVS (chorionic villus sampling)
 defined, 1:258, 2:97
 uses, 3:184, 3:187
- Cyanobacteria
 as chloroplast ancestor, 2:195, 3:57
 in stromatolites, 2:21–22, 2:22
- Cycle sequencing. *See* Automated sequencers
- Cyclic AMP response element binding protein (CREB), signal transduction role, 4:88, 4:89
- Cyclic AMP, signal transduction role, 1:5–6, 1:110, 4:89
- Cyclin-dependent kinases (CDKs), 1:104, 1:105–108, 1:106
- Cyclins, cell cycle role, 1:105–108
- Cyclosporine, 4:141
- Cystic fibrosis (CF), 1:199–203
 CFTR protein, 1:201–202, 2:127–128
 clinical features, 1:199–200, 2:202
 damaged DNA role, 1:239
 and diabetes, 1:212
 “Disease Model” mouse strains, 3:62
 disease prevalence studies, 1:214
 gene therapy, 2:81
 genetic mutations, 1:200–202, 1:200
 genetic testing, 1:202–203, 2:168, 3:177, 3:186
 inheritance patterns, 1:200–201, 1:213, 2:127–128, 2:203, 3:142, 3:193–194
 laboratory diagnosis, 1:200
 pleiotropy, 2:206
 preimplantation genetic diagnosis, 3:186
 Punnett squares to predict, 3:193–194, 3:194
 as simple trait, 1:177, 2:101–102
 statistics, 1:199
 transmembrane conductance regulator gene, 4:127
 treatment, 1:201, 1:201
- Cystic fibrosis transmembrane conductance regulator (*CFTR*) gene, 1:202, 2:60, 2:127–128
- Cystine (Cys)
 chemical properties, 3:200
 genetic code, 2:85, 4:137
 in transcription factors, 4:115
- Cytochemists
 defined, 2:187
in situ hybridization role, 2:187–189
- Cytochrome *b* protein, 2:194
- Cytochrome C protein
 apoptosis role, 1:33
 evolution of, 3:94
- Cytochrome P450, poor and ultrarapid metabolizers, 3:144–145
- Cytogenetic analysis
 chromosomal banding techniques, 1:125–129
 defined, 1:119, 2:44, 3:10
 for Down syndrome, 1:257–258
- Cytogeneticists, defined, 3:182
- Cytokines
 from cloned genes, 1:153
 defined, 4:75, 4:85
 gene paralogs, 2:158
 signal transduction role, 4:85
- Cytokinesis
 defined, 1:113, 3:29
 lacking in polyploidy, 2:69
 in meiosis, 3:29
 in mitosis, 3:59
- Cytology/cytologists, defined, 1:130, 3:21
- Cytomegalovirus, 1:30, 3:124
- Cytoplasm, 1:111
 defined, 1:109, 2:53, 3:29, 4:22
 of eubacteria, 2:12
- Cytosine, 1:216
 concentration, for *in situ* hybridization, 2:187
 depyrimidination, 1:240
 in DNA alphabet, 2:83, 4:106
 and DNA structure, 1:215–220, 1:250–251, 2:50, 2:51, 3:94
 evolution of, 2:22–25
 methylation of, 2:184, 3:46–49, 3:47, 3:95, 3:100
 mutagenic base analogs, 3:87
 origin of term, 1:249
 protonated, 1:218
 in RNA alphabet, 4:46–47, 4:106
 structure, 3:115, 3:116, 3:118–119, 3:118, 3:119, 4:48, 4:49
See also Base pairs
- Cytoskeleton
 actin protein role, 2:29
 signal transduction role, 4:87
 structure, 1:113
- Cytosol
 defined, 1:109
 glucose breakdown, 1:111
 ribosomes in, 4:43

D

- D4 and D5 dopamine receptor genes (*DRD4* and *DRD5*), 1:41
- DAG, signal transduction role, 4:89
- DAPI dyes, 1:127
- Darwin, Charles, 1:129, 1:131, 2:17, 2:26, 2:156, 4:67, 4:92
- Darwinian evolution (Darwinism), 1:131, 2:17, 2:26, 2:156, 3:32, 4:67, 4:92
- DAT* (dopamine transporter) gene, 1:41
- Data Deficient population status, 1:186
- Databases
 analysis, 1:55–56
 bioinformatics data, 2:124
 of chromosomal abnormalities, 3:182
 clinical trials, 1:102–103

- data acquisition, 1:54
 data integration, 1:56
 development, 1:54
 disease registry, 1:102
 of DNA samples, 3:192–193
 of gene and protein sequences, 1:52–54, 2:123–124, 2:156, 2:173, 2:174, 2:176, 3:18, 3:198, 3:208–209
 gene mining companies, 2:123–124
 high-throughput screening data, 2:150
 Internet tools, 2:212
 relational, 1:55
 twin registries, 4:84, 4:159–161
- Databases, specific**
 BLOCKS, 3:209
 GenBank, 1:52, 1:54, 2:156, 2:174
 Human PSD, 3:209
 Pfam, 3:209
 PRINTS, 3:209
 PROSITE, 3:209
 SWISS-PROT, 1:52, 3:209
 Yeast Proteome, 3:209
- DAX1* gene, 4:81
 DBH mice, 1:5
 DCMP (deoxycytosine monophosphate), 3:117
 DDAPV, to treat hemophilia, 2:145
 De novo mutations, 1:120
 De Silva, Ashanti, 4:77–78
 De Vries, Hugo, 1:130, 3:102
 Dead Sea, Archaea adaptations, 1:37
 Deafness. *See* Hearing loss
Deamination
 errors, DNA damage, 1:240–242, 1:240, 3:47, 3:88
 of methyl cytosine, 3:47, 3:47
 Death, as biological process, 1:31
 Death, cell. *See* Apoptosis
 Decker, Donovan, 2:75
 DeCODE, populations studies, 2:124
 Defective particles, viruses, 4:119
 Degeneracy of genetic code, 2:27, 2:84, 4:4, 4:5, 4:135
 Dehydroepiandrosterone sulfate, and breast cancer, 1:90
Deinococcus radiodurans, genetic studies, 1:139–140
 Delano, Susana, 3:71
 Delbrück, Max, 1:192, **1:203–204**, 3:103, 4:172
 Deleterious, defined, 2:147, 2:190, 2:191, 3:89, 4:175
 DeLisi, Charles, 2:173
 Delta-retrovirus, characteristics, 4:35
Dementia
 aging and, 1:7
 defined, 1:1, 1:7, 1:15, 3:189
 non-Alzheimer causes, 1:15
 progeria and, 1:1
 See also Alzheimer's disease; Brain disorders; Psychiatric disorders
Demography, defined, 1:191
Denaturation
 in cycle sequencing, 1:198
 defined, 2:47
 DNA unwinding, 1:218
 in PCRs, 3:157
Dendritic cells, immune system function, 2:152
Dentatorubral-pallidolysian atrophy, triplet repeat diseases, 4:150
 DeoxyATP, SCID role, 4:75
 Deoxycytosine monophosphate (dCMP), 3:117
 Deoxynucleotides, defined, 1:230
 Deoxyribonuclease enzymes, function, 3:112, 3:113–114
 Deoxyribonucleic acid. *See* DNA
 Deoxyribose sugars
 discovery, 1:249
 and DNA structure, 1:215–220, 2:50, 2:51
 structure, 3:115, 3:116, 3:117
Department of Education, genetic discrimination studies, 2:94
Department of Energy (DOE)
 Human Genome Project role, 2:173
 radiation-caused-mutation studies, 2:171, 2:173
Depurination, 1:240
Depyrimidination, 1:240
 DeSalle, Rob, 1:190
 Desipramine, for ADHD, 1:39
 Detergents, genetically engineered proteases, 1:63–64, 1:73
 Detoxifying agents, to treat metabolic diseases, 3:39, 3:45
 Deuteranopia/anomaly, 1:172
Development, defined, 1:204
Development, genetic control of, 1:204–209
 apoptosis role, 1:208–209
 axis determinations, 1:206–207, 2:68
 DNA methyltransferase role, 3:49
 early studies, 3:103
 embryonic stem cells, 2:3–6
 European and American models, 1:205–206
 in fruit flies, 2:44–45
 gene expression regulation, 2:65–66
 hedgehog protein role, 3:181
 homeobox sequence, 1:208
 homeotic genes, 1:207–208, 2:29, 2:65–66
 hormonal regulation, 2:159–164
 McClintock's hypothesis, 3:21–22
 mitosis and, 3:57
 morphogen gradients, 1:206
 polyploids, 3:164, 3:166
 in roundworms, 1:204, 1:205, 1:206, 1:208, 4:63–64
 transcription factor role, 1:205, 4:115
 transgenic animals to study, 4:125–127
 triplet repeats and, 4:149
 in zebrafish, 3:61, 4:182–183
 See also Imprinting; Sex determination; Sexual development
DHPLC (denaturing high-performance liquid chromatography), 2:165–167
DHT (dihydrotestosterone), function, 2:163
Diabetes, 1:209–212
 A3243G mutations, 3:54
 and birth defects, 1:77, 1:82
 and cardiovascular disease, 1:101, 1:209
 as complex trait, 1:178, 1:210, 1:213, 2:102
 congenital generalized lipodystrophic (CGL), 1:211, 2:203
 diet and, 1:210
 “Disease Model” mouse strains, 3:62
 embryonic stem cell research tools, 2:5–6
 excess blood glucose, impact, 1:209
 familial partial lipodystrophic (FPLD), 1:211
 genetic counseling, 2:90
 genetic susceptibility to complications, 1:212
 gestational (GDM), 1:209, 1:212
 maternally inherited diabetes and deafness (MIDD), 1:211
 maturity onset (MODY), 1:211
 as metabolic disease, 3:42
 nongenetic, 1:210
 physical activity and, 1:210
 populations studies, 2:124
 racial and ethnic differences, 1:210
 statistics, 1:209

- Diabetes (continued)
 thiamine-responsive megaloblastic anemia syndrome (TRMA), 1:212
 transient neonatal (TNDM), 1:212
 type 1 (T1DM), 1:209–210, 2:202
 type 2 (T2DM), 1:209, 1:210, 1:212, 3:125, 3:154
 Wolfram syndrome, 1:212
See also Insulin
- Diagnostic and Statistical Manual (DSM-IV)*, 1:4, 3:213
- Diagnostic testing kits, genetically engineered products, 1:64
- Diagnostic tests
vs. population screening, 3:175
 recombinant DNA tools, 4:6
- Diamond v. Chakrabarty*, 1:72, 3:136–137
- Diastrophic dysplasia, linkage disequilibrium role, 2:38
- Dicer enzymes, 4:53, 4:54
- Dichromat color vision defects, 1:171, 1:172
- Dicots, genetic modification techniques, 2:107–108
- Dictyostelium discoideum* (slime mold), as model organism, 3:60
- Dideoxy method, 4:64
- Dideoxynucleotides, 1:43, 1:198, 1:199
- Diet. *See* Nutrition
- Diethylstilbestrol, and breast cancer risk, 1:90
- Diffuse B-cell lymphoma (DLBCL), chemotherapy, 3:146
- Digestive enzymes, processing of, 3:204
- Dihydrotestosterone (DHT), function, 2:163
- Dimerize, defined, 2:164
- Dimethylsulfate, as mutagen, 1:242
- Dimorphisms
 of Alu sequences, 4:145–146, 4:146
 defined, 4:177
See also Polymorphisms, DNA
- Dimples, facial, 2:191, 2:202
- Diploid
 accumulation of recessive mutations, 3:93
 defined, 2:27–28, 3:163, 4:176
 somatic cells as, 3:24, 3:60
 virions as, 4:35
 zygotes as, 3:24
- Direct DNA absorption, 4:133
- Directive on Data Protection, 3:192
- Disabled individuals, legal definitions of, 2:93
- Disaccharides, defined, 3:132
- Disaster victims, DNA profiling to identify, 1:234, 1:238
- Disease concordance, defined, 1:214
- “Disease Model” mouse strains, 3:62
- Disease registry databases, 1:102
- Disease-resistant animals, transgenic, 2:106
- Disease-resistant crops
 genetically engineered, 1:63, 2:108
 genomics role, 2:121
 nematode resistance, 1:149
 statistics, 1:57–58
 transgenic, 2:106, 2:107, 4:134
- Diseases
 as bioterrorism tools, 1:69–70
 genetically engineered testing tools, 1:64
- Diseases, genetics of, 1:213–215
 allele scoring strategies, 2:192
 antisense nucleotide tools, 1:29, 1:30–31
 chaperone role, 1:118
 clinical geneticist role, 1:149–151
 complex traits, 1:177–181
 DNA methylation role, 3:48, 3:49
 DNA microarray analysis, 1:227–229
E. coli as research tool, 2:10–11
 early studies, 1:250
vs. environmental factors, 1:213, 1:214
 epidemiologist role, 2:7, 2:56–57
 epistatic interactions, 2:8–9
 extranuclear inheritance, 2:194, 2:198
 eye color changes, 2:33
 familial aggregation studies, 1:213–214
 frameshift mutation role, 4:4–5
 gene discovery, 2:57–61
 gene targeting tools, 2:73
 and gene therapy effectiveness, 2:74
 gene-environment interactions, 1:213, 1:214, 2:55–57
 genetic counseling, 2:87–91, 2:91–92
 genomic medicine, 2:118–120, 2:121–122
 genomics industry role, 2:124–125
 growth disorders, 2:129–132
 Hardy-Weinberg predictions, 2:135
- heterozygote advantage, 2:146, 2:148
- HPLC to study, 2:166–167
- Human Genome Project contributions, 1:214–215
- imprinting defects, 2:183, 2:185
- inbreeding role, 2:189–191, 3:174
- linkage and recombination tools, 3:4–8, 3:26
- and longevity, 1:6–8
- Mendelian and complex disorders, 1:213
- Mendelian inheritance, 3:36, 3:37
- molecular biologist role, 3:70
- molecular chaperones, 1:118
- mouse models, 3:62
- mutation rates, 3:99–100
- mutation role, 1:239, 3:93
- pedigrees as tools, 3:7, 3:36, 3:138–142
- pharmacogenomics research, 3:146–147
- polymorphisms as tools, 3:162
- population screening, 1:202–203, 2:6–7, 2:99–100, 2:143, 3:175–178
- prokaryote chromosome mapping, 1:139–140, 1:142
- public health role, 3:216–219
- quantitative traits, 4:1–3
- relative risk ratios, 1:214
- rodent models to study, 4:61–62
- stem cell research tools, 2:5
- transgenic animal tools, 4:124, 4:127
- twin and adoption studies, 1:214
- unequal crossovers and, 1:197
- X-chromosome inactivation, 3:77–80, 4:175–176
- See also* Dominant disorders, autosomal; Human disease genes, identification of; Metabolic diseases; Mitochondrial diseases; Recessive disorders, autosomal; *specific diseases*
- Diseases, of plants
 crown gall, 1:9, 1:35, 4:132
 fire blight, 2:107
Phytophthora infestans, 2:108
 viroids, 4:162–163
 viruses, 4:165, 4:168
 virusoids, 4:164
See also Disease-resistant crops
- Disjunction, 1:120
- Dispersion, in statistical analyses, 4:96
- Disulfide bridges, 3:200, 3:201
- Dizygotic, defined, 4:83

- Dizygotic twins. *See* Twins, fraternal
- DLBCL (diffuse B-cell lymphoma), chemotherapy, 3:146
- D-loop, of mtDNA, 3:53
- DM locus-associated homeodomain protein (DMAHP), 4:150
- DMD. *See* Muscular dystrophy, Duchenne
- DMPK gene, 3:86, 4:150
- DNA (deoxyribonucleic acid)
- alphabet, 2:83, 4:106
 - polymer formation, 3:117–118, 3:118
 - preserving, *in situ* hybridization, 2:188
 - of viruses, 4:165–168
- DNA, alphoid, 1:146
- DNA, chloroplast (cpDNA)
- evolution of, 2:194–195
 - multiple copies, 2:196
 - as nuclear markers, 2:198
 - vegetative segregation, 2:197
 - See also* Chloroplasts, genome
- DNA, complementary (cDNA)
- defined, 1:224
 - as gene cloning tool, 1:153, 1:154
 - libraries, 1:157, 1:223, 1:224
 - as pseudogenes, 3:211
 - recombinant DNA tools, 4:6
 - in reverse-transcription reactions, 3:159, 4:40
 - in situ* hybridization role, 2:187–188
- DNA damage. *See* DNA repair; Mutagens; Mutations
- DNA, exogenous, 4:121–123
- DNA fingerprinting. *See* DNA profiling
- DNA footprinting, 1:220–221
- applications, 1:221, 4:114
 - how it works, 1:220, 1:221
- DNA glycosylase enzyme, 1:242, 1:243, 1:244
- DNA gyrase enzyme, 2:15, 4:14–15, 4:18
- DNA helicase enzyme
- replication role, 4:12, 4:13, 4:13, 4:14, 4:18
 - and Werner's syndrome, 1:3, 4:14
- DNA libraries, 1:222–225
- applications, 1:222
 - artificial chromosome tools, 1:222
 - cDNA, 1:157, 1:223, 1:224, 4:6
 - combinatorial chemistry role, 1:175
 - expression libraries, 1:224, 4:6
 - gene cloning tools, 1:155
 - of gene mutations, 2:122, 4:182–183
 - genomic libraries, 1:222, 1:224
 - of oligonucleotides, 1:176–177
 - recombinant DNA tools, 4:6
 - SAGE, 1:229
- DNA ligase enzyme
- DNA repair role, 1:244
 - DNA replication role, 4:16, 4:18
 - gene cloning role, 1:152, 1:154, 4:128
 - recombinant DNA tool, 4:5, 4:33
- DNA, linker, 1:133
- DNA methylation. *See* Methylation, of DNA
- DNA methyltransferases (DNMTs), 3:46, 3:49
- DNA microarrays, 1:225–229, 1:226
- Affymetrix, 1:226
 - applications, 1:228
 - bioinformatics tools, 1:227
 - classification analysis, 1:228
 - clustering analysis, 1:227
 - expression analysis, 1:225, 1:228, 4:40
 - as genomic medicine tool, 2:119
 - high-throughput screening tools, 1:175, 2:150
 - molecular hybridization tools, 1:225–226
 - as pharmacogenomics tool, 3:146
 - printed, 1:226
 - as proteomics tool, 2:49
 - SAGE analysis, 1:228–229
 - to study quantitative traits, 4:3
- DNA, mitochondrial (mtDNA)
- characteristics, 3:55–56
 - D-loop, 3:53
 - in eggs and sperm, 2:197, 3:51
 - evolution of, 2:194–195, 3:57
 - as evolutionary study tool, 3:66–67
 - heavy and light strands, 3:56
 - multiple copies, 2:196
 - mutations, 3:51, 3:53–54
 - as nuclear markers, 2:198
 - sequence comparisons, 3:57
 - transcription of, 3:56
 - vegetative segregation, 2:197
 - See also* Mitochondrial genome
- DNA polymerase I, 4:16, 4:17, 4:18
- DNA polymerase III
- in prokaryotes, 4:13, 4:16–17, 4:18
 - proofreading, 4:17, 4:18
 - sliding clamps, 4:16, 4:17, 4:18
- DNA polymerases, 1:230–233
- α , 1:231, 4:17, 4:18
 - β , 1:232
 - σ , 1:231–232, 4:17
 - ϵ , 1:231–232, 4:17
 - γ , 1:232
 - σ , 4:17
 - in cycle sequencing, 1:199
 - DNA-dependent, 4:168
 - E. coli*, 1:231, 3:157
 - end-replication problem, 4:104
 - as gene cloning tools, 1:152–153, 1:153, 1:154
 - human, 1:231–232, 4:41
 - incorrect polymerization, 1:231, 1:232, 3:113–114
 - mechanisms, 1:230–231, 1:231
 - in PCR, 3:154–159
 - polymerization function, 1:230
 - produced by cloning, 1:154
 - repair function, 1:230, 1:232, 1:241, 1:244, 1:248, 3:113–114
 - shape, 1:232–233
 - size range, 1:232
 - Taq, 1:38, 1:199, 3:157
 - See also* Transcriptase, reverse
- DNA polymorphisms. *See* Polymorphisms, DNA
- DNA primase enzyme, 4:13, 4:16, 4:18
- DNA primers, 1:64, 4:71
- DNA profiling (fingerprinting), 1:233–239
- blood typing, 1:233–234
 - CODIS, 1:235, 1:236, 1:237
 - contamination concerns, 1:235
 - DNA polymorphisms, 1:234–235, 3:163
 - DNA purification tools, 3:220
 - and DNA stability, 2:84
 - history, 1:73
 - legal concerns, 3:3, 3:4
 - matching techniques, 1:236, 1:238
 - misuse of statistics, 1:238
 - mtDNA D-loop as tool, 3:53
 - PCR as tool, 3:154, 3:156
 - privacy concerns, 3:192–193
 - repeated sequences as tools, 1:235–236, 4:11
 - restriction enzyme tools, 4:32, 4:33–34
 - uses, 1:187–189, 1:233, 1:234, 1:238–239
 - Y chromosome analysis, 4:178
- DNA purification, 3:220–221
- electrophoresis techniques, 2:49–50

- DNA purification (continued)
 in molecular anthropology, 3:68
 procedures, 3:220–221
 uses, 3:220
- DNA, recombinant, 4:5–7
 in Alzheimer's disease studies, 1:16
 as biotechnology tool, 1:9–11, 1:63, 2:107, 4:6–7
 conjugation and, 1:142, 1:185, 2:15
 defined, 1:63, 4:5
 drugs derived from, 1:69, 1:72, 2:125, 4:6–7
 ethical issues, 3:152, 4:5–7
 fruit fly transposons, 2:43
 for gene cloning, 1:152–153
 as gene therapy tool, 4:5–6
 as mapping tool, 1:141–142, 3:7, 4:6
 molecular biologist role, 3:71
 patents on, 3:137–138
 for PCR, 3:157
 plasmid vectors, 3:152
 research history, 1:71, 4:5
 restriction enzyme tools, 4:31, 4:33
 rodent models, 4:61
 in selectable markers, 3:17
 techniques, 4:5–6
 for vaccines, 1:253, 4:6
 See also Restriction enzymes
- DNA repair, 1:239–248
 apoptosis and, 1:32, 1:33
 bacteriophage replication and, 4:118–119
 BER pathway, 1:240, 1:242–244, 1:243
 cytochrome P450s, 3:89
 deoxyribonuclease enzyme role, 3:113–114
 DNA mismatch repair, 1:246–248, 3:114
 double-strand breaks, 1:248
 future research, 1:248
 helicase and, 1:3
 in mitochondria, 3:79
 mutation rate role, 3:100
 mutations, role in creating, 3:95, 3:100
 nucleotide excision repair, 1:244–246, 1:245
 polymerases and, 1:230, 1:232
 proofreading exonucleases, 1:231, 1:232, 3:113–114, 4:17, 4:18
 recombinational, 1:248
 regulation of, 1:105
 of Y chromosomes, 4:177
- DNA repair genes, mutations, 1:95, 1:99, 1:169, 1:241–242
- DNA repetitive sequences, 4:7–12
 applications, 4:11
 concatamers, 4:118
 in eukaryotic cells, 1:132
 gene evolution role, 2:28, 2:114–115, 4:9, 4:10–11
 in higher eukaryotes, 2:115
 in human genome, 4:7
 in kinase gene, 3:153–154
 in prokaryotes, 2:117
 retropseudogenes, 4:10
 STRPs, 3:6, 3:14
 in telomeres, 4:8–9, 4:10, 4:104, 4:105
 VNTRs, 1:236, 3:6, 3:14, 3:162, 4:8–9
 See also Alu sequences; Short tandem repeats (microsatellites); Transposable genetic elements; Triplet repeat diseases
- DNA replication, 4:12–19
 base substitution mutations, 3:95
 defined, 1:19, 2:21, 4:5
 directionality, 3:117–118, 3:155–156, 4:12, 4:15–16, 4:15, 4:135
 DNA coiling in nucleus, 3:122
 early studies, 1:251–252
 energy requirements, 1:111
 impact of mutagens, 1:19
 leading and lagging strands, 4:14, 4:15–16, 4:15
 nuclear envelope breakdown and re-forming, 3:59, 3:120
 Okazaki fragments, 4:14, 4:15, 4:16, 4:17, 4:18
 by plasmids, 3:151–152
 recombinant DNA, 4:5–6
 regulation of, 4:17–18
 replication fork (unwinding), 4:13–15
 of retroviruses, 4:36–37, 4:37
 as semiconservative, 4:12
 See also Cell cycle; DNA polymerases; Meiosis; Mitosis; Transcription
- DNA replication, eukaryotes
 Cdc6, 4:18
 distinguished from prokaryotes, 4:17
 DNA polymerases, 4:17
 Dna2 protein, 4:17
 duration, 4:12
 Fen1 protein, 4:17
 MCM proteins, 4:18
 ORC, 4:13, 4:13
- origins of, 4:12–13
 origins of DNA replication, 4:13
 pre-replicative complexes, 4:18
 RNase H, 4:17
 single-strand binding protein (RPA), 4:14
 of telomeres, 4:17
 topoisomerase, 4:14–15, 4:18
- DNA replication, prokaryotes
 distinguished from eukaryotes, 4:17
 DNA gyrase, 4:14–15, 4:18
 DNA polymerase III, 4:13, 4:16–17, 4:18
 dnaA protein, 4:13, 4:18
 duration, 4:12
 origins of, 4:13
 primers, 4:16
 replication fork (unwinding), 4:14, 4:14
 single-strand binding protein (SSB), 4:13, 4:14, 4:18
- DNA, selfish. *See* Transposable genetic elements
- DNA sequencing, 4:69–74
 bioinformatics tools, 1:52–56, 2:29
 BLAST applications, 3:196
 blotting procedures, 1:86–89, 1:152
 chain termination method, 1:43, 1:198, 4:71–73, 4:72
 computational biologist role, 1:181
 to confirm genetic code accuracy, 2:86
 cycle sequencing technique, 1:198–199
 cystic fibrosis gene, 1:201–202
 databases, 1:52–54, 2:123–124, 2:156, 2:173, 2:174, 2:176, 2:212
 defined, 1:35
 to detect SNPs, 3:162
 DNA library tools, 1:222–224
 DNA purification tools, 3:220–221
 electrophoresis tools, 2:49–50
 with fluorescent dyes, 4:72–73, 4:73
 footprinting to study, 1:221
 gene cloning tools, 1:152, 1:154–157
 genomics industry role, 2:123–125
 genomics role, 2:120–122
 high-throughput screening tools, 2:150
 homeobox sequence, 1:208

- as indicator of evolutionary relatedness, 2:156–158
of insulin, 4:64
Internet tools, 2:212
mass spectrometry tools, 3:18–20
Maxam-Gilbert method, 4:70–71, 4:71
as molecular anthropology tool, 3:63–65
pairwise distributions, 3:167–168, 3:168
palindromic, 1:218
patent issues, 3:137
procedures, 3:1
Rh polypeptide, 1:86
in situ hybridization tools, 2:186–189
See also Automated sequencers; Human Genome Project; Mapping genes
- DNA sequencing, specific organisms
Arabidopsis thaliana, 2:121, 2:172
E. coli, 1:140, 2:10, 2:11, 2:86, 2:116, 2:172
eubacteria, 2:172
fruit flies, 2:172
model organisms, 2:172
rice, 2:172
rodents, 3:61–62, 4:60
roundworms, 2:121, 2:122, 2:172, 2:174, 4:62–63
yeasts, 2:121, 2:172
- DNA structure, 1:215–220, 2:51
3' and 5' ends, 1:216, 1:216, 2:51, 3:115, 3:116, 3:117–118, 3:118, 3:155–156, 4:12, 4:15–16, 4:15
antiparallel, 1:217
antisense nucleotides, 1:30–31, 1:30
base stacking, 1:217
B-DNA (canonical), 1:215–218, 1:218, 1:219
coiled in nucleus, 3:121, 3:122
components, 1:215–216, 3:115, 3:116, 3:117, 4:114
cruciform, 1:218, 1:219
curved, 1:219, 1:220
distinguished from RNA, 4:46
genomics role, 2:120–122
glycosidic bonds, 1:216, 1:216
hydrophobic interactions, 1:217
intermolecular triplex, 1:218, 1:219
intramolecular triplex, 1:219, 1:219
left-handed Z, 1:219–220, 1:219
looped domains, 1:126, 1:134, 1:134, 2:64, 2:65, 4:114
major and minor grooves, 1:217, 1:217, 1:218, 4:115, 4:116
packaged in nucleosomes, 4:111
phosphodiester bonds, 1:216, 1:216, 1:244
quadruplex, 1:219, 1:219
slipped-strand, 1:218, 1:219
stability, 2:84
unwound, 1:218, 1:219
van der Waal's forces, 1:217
See also Base pairs; Deoxyribose sugars; Double-helix DNA model; Hydrogen bonds; Nucleotides
- DNA structure and function, history, 1:248–253
DNA as transforming factor, 1:250–251, 1:250, 3:103, 3:104, 3:105, 4:120, 4:124
DNA discovery, 1:215, 1:249–250, 2:112, 3:103
DNA replication studies, 1:251–252
Watson and Crick's model, 1:62, 1:192–193, 1:251, 2:50, 2:120, 2:171, 3:105, 4:172
See also Chromosomal theory of inheritance, history; Gene nature, history
- DNA thermal cyclers, 3:157
DNA, transformed (T-DNA), 1:35
DNA transposons, 4:8, 4:9–10, 4:11
DNA triplets, as universal, 1:61, 1:152
DNA uptake, 4:121
DNA vaccines, 1:253–255
advantages, 1:254
clinical studies, 1:255
development, 2:121
testing on cattle, 1:254
uses, 1:72, 1:255
vaccination techniques, 1:254–255
- Dna2 protein, 4:17
DnaA protein, 4:13, 4:18
DnaK chaperones, energy requirements, 1:117
DNase
cystic fibrosis treatment, 1:201
DNA footprinting role, 1:220
in DNA purification, 3:221
as indicator of transformation, 4:123
DNMTs (DNA methyltransferases), 3:46, 3:49
Dobzhansky, Theodosius, 3:76
DOE (Department of Energy)
Human Genome Project role, 2:173
radiation-caused-mutation studies, 2:171, 2:173
Dolly the sheep, 1:158, 1:160, 1:162, 1:163–164
Dominance hypothesis, for heterozygote advantage, 2:147
Dominant alleles
codominance, 2:200
defined, 3:194, 4:69
distinguished from epistasis, 2:7
distinguished from imprinted genes, 2:183–184
genotype role, 2:127, 2:127–128
heterozygote advantage role, 2:147–148
incomplete dominance, 2:200, 3:36
Mendel's hypotheses, 1:147–148, 3:31, 3:33, 3:35–36
molecular meaning of, 2:200–201
See also Inheritance patterns
- Dominant disorders, autosomal
achondroplasia, 1:75–76, 2:130
Alzheimer's disease, 1:16–18
Beckwith-Wiedemann syndrome, 2:132, 2:185
cartilage matrix protein mutations, 2:132
defined, 3:153
diabetes, familial partial lipodystrophic, 1:211
diabetes, maturity onset, 1:211
endocrine disorders, 2:129–130
familial adenomatous polyposis, 1:168–169
familial hypercholesterolemia, 2:57, 2:201, 2:202
gain-of-function mutations, 3:96
gene therapy effectiveness, 2:74
Huntington's disease, 2:103–104, 2:127, 2:201
inheritance patterns, 1:213, 2:201, 2:205
Marfan syndrome, 1:75, 1:76
muscular dystrophies, 3:85–86, 3:153–154
neurofibromatosis, 3:153
progeria, 1:1–3
psychiatric disorders, 3:214
retinoblastomas, 4:153
selection for/against, 4:69
spinocerebellar ataxia, 2:205
table of, 2:202
toxic-gain-of-function mutations, 2:201
Dominant disorders, X-linked, 1:130–131, 2:203, 3:73–74, 3:91–92, 4:175
Dominant negative effect, 2:201

- Donahue's syndrome, progeroid aspects, 1:2
- Donor insemination, 4:24
- Donors
- blood, 1:83
 - gametes, privacy of, 4:28
 - organ, primates as, 4:142
 - tissue, legal concerns, 3:3–4
 - transplant, defined, 4:139
- Dopamine
- ADHD and, 1:39–41, 1:41
 - neurotransmitter systems, 1:5
- Dopamine transporter (*DAT*) gene, 1:41
- Dosage compensation, defined, 3:81
- The Double Helix* (J. Watson), 4:173
- Double-helix DNA model,
- 1:216–218, 1:218, 2:50, 2:51, 3:118–119
 - disrupted by mutagens, 3:88, 3:94
 - distorted by transcription factors, 4:112, 4:115
 - distortion, repairing, 3:114
 - right-handed helix, 1:217
 - unwinding in DNA replication, 4:12–16
 - of Watson and Crick, 1:62, 1:192–193, 1:251, 1:252, 3:105, 4:172
- DoubleTwist, bioinformatics, 2:124
- Down, J. Langdon, 1:256
- Down syndrome, 1:256–258
- association with leukemia, 4:155
 - average IQ, 1:79
 - chromosomal basis, 1:79, 1:256–257, 3:29, 3:97, 3:110, 3:111, 3:164
 - clinical features, 1:3, 1:256, 1:257, 3:110
 - cytogenetic analysis, 1:257
 - history, 1:119, 1:256
 - mosaic, 1:257
 - prenatal genetic testing, 1:258
 - progeroid aspects, 1:2–3
 - senescence, 1:3
 - statistics, 1:75
 - translocation, 1:257
 - trisomy 21, 1:257, 3:29
- Down syndrome, prenatal genetic testing, 2:97
- DRD2 mice, 1:5
- Drosophila*. *See* Fruit flies
- Drug dependence
- and birth defects, 1:77, 1:81
 - defined, 1:4
 - genetic components, 3:214
 - See also* Addiction
- Drugs (medications)
- acceptable toxicity, 4:171
 - Ames test, 1:20–21
 - biopharmaceuticals, 2:125, 4:129
 - and birth defects, 1:80, 1:206
 - chemotherapy, 3:146, 4:74
 - combinatorial chemistry to synthesize, 1:175, 1:176–177
 - developing new, 2:121, 2:124–125, 3:206
 - estrogen replacement, 1:90
 - fertility, 4:19, 4:21–22, 4:28
 - high-throughput screening tools, 2:149–150
 - how they work, 2:149, 3:206, 4:44
 - and hyperpigmentation of iris, 2:33
 - immunosuppressive, 4:141
 - pharmaceutical scientist role, 3:142–144
 - poor and ultrarapid metabolizers, 3:144–145
 - side effects, preventing, 2:118–120, 2:122, 3:144–146, 4:22
 - signal transduction targets, 4:91
 - See also* Antibiotic resistance; Antibiotics; Pharmacogenetics and pharmacogenomics
- Drugs (medications), for specific diseases
- Alzheimer's disease, 1:181
 - cancer, 1:97, 1:153, 1:170
 - familial hypercholesterolemia, 2:57
 - hemophilia, 2:145
 - HIV/AIDS, 2:150–151, 2:155–156, 4:41
 - muscular dystrophies, 3:87
 - SCID, 2:81
 - sickle-cell disease, 2:138
 - Tay-Sachs disease, 4:102
 - viruses, 4:171
- DSM-IV (*Diagnostic and Statistical Manual*), 1:4, 3:213
- DsRNA. *See* RNA, double-stranded
- Duchenne de Boulogne, 3:84
- Dunker population, inbreeding coefficient, 2:190
- Dunnigan-type lipodystrophy, 3:125
- Dupont company, Genesis 3000 sequencer, 1:44
- Duty of candor, for patents, 3:138
- Dwarfism
- achondroplasia, 1:75–76, 2:82
 - prenatal genetic testing, 3:186
 - as quantitative trait, 4:1
 - skeletal dysplasias, 2:130–132
- Dyskeratosis congenita, progeroid aspects, 1:2
- Dystrophia myotonica. *See* Muscular dystrophy, myotonic
- Dystrophin* gene, 2:77–78, 3:84–85, 3:100
- ## E
- E. coli* bacterium. *See* *Escherichia coli*
- E2F family transcription factors, 1:106, 1:107
- Ear wax consistency, 2:192
- Earlobes, free, 2:191
- Ears, hairy, 2:192
- Ecological (sympatric) speciation, 4:93
- Ecology. *See* Conservation biology, genetic approaches
- Ecosystem, defined, 1:59, 4:130
- Ectopic expression, defined, 2:74
- Ectopic pregnancies, 4:22
- Edman degradation technique, 3:197
- Edman, Pehr, 3:197
- EDTA (ethylenediaminetetraacetic acid), 3:221
- Educators, 2:1–3
- genetic counselors as, 2:88–90, 2:91–92
 - medical role, 2:90
- EGFR* gene (Epidermal Growth Factor Receptor), 1:95
- Egg donation, assisted reproduction, 4:24, 4:28
- Eggs
- chromosomal aberrations, 1:121
 - developmental processes, 2:34–35, 3:29
 - enucleated, cloned organisms, 1:158, 1:162–164
 - haplodiploidy, 2:113
 - as haploid, 2:113, 2:115, 3:24, 3:60, 3:163
 - human, size, 1:108
 - meiosis and, 1:115, 2:33–34, 3:24–29, 3:60, 3:75
 - microassisted fertilization, 4:22
 - mRNA stability, 4:51
 - mtDNA in, 2:197, 3:51
 - mutations, 2:192, 3:99
 - zona pellucida, 2:34, 2:35
 - See also* Fertilization
- Eigsti, O.J., 3:165
- Electron microscopy, 4:164–165
- Electron transport chain (ETC)
- coding, mitochondrial *vs.* nuclear genomes, 3:52–53, 3:52
 - mitochondrial role, 1:111, 2:194, 3:41–42, 3:51–53, 3:52

- Electrophoresis
 capillary, 4:74
 defined, 1:198
 See also Gel electrophoresis
- Electroporation, 2:107, 4:123, 4:133
- Electrospray ionization (ESI), 3:20, 3:208
- Ellis-van Creveld syndrome, 3:174
- Elongation factors, translation, 4:138
- Eluting, defined, 2:165
- EMBL database, 1:52
- Embryonic development. *See* Development, genetic control of
- Embryonic stem cells. *See* Stem cells, embryonic
- Embryos, human
 cloning, 1:69, 1:158–160, 1:161
 eight-cell (blastocyst) stage, 1:163, 4:20, 4:23
 frozen, 4:25, 4:26
 genetic selection of, 4:29–30
 personhood of, 4:27–28, 4:29
 twins, 2:35–36, 4:155–156
 See also Reproductive technology
- Embryogenesis, defined, 4:182
- Emergenesis, defined, 2:9
- Emerin proteins, 3:125
- Emerson, R.A., 3:10
- “Emia,” as suffix, 3:41
- Employee Retirement Income Security Act (ERISA), 2:93–94
- Employment discrimination, genetic testing and, 1:18, 1:68, 2:92–94, 2:100, 2:177, 3:4, 3:193
- EMS (ethyl methanesulfonate), 1:19
- Endangered species
 cloning, 1:158, 1:161, 1:164
 endangered population status, 1:186, 1:190
 evolutionary relationship determinations, 1:188, 1:189
 extinction vortex, 1:188
 genetic diversity, 1:187, 3:167
 illegal trading in, 1:186, 1:189–190
 inbreeding and, 2:190
 population bottlenecks, 1:188–189
 relocation and reintroduction, 1:186–187, 1:190
 threatened population categories, 1:186
 viability analyses, 1:187
- Endochondral ossification, 2:130
- Endocrine cells, hormone secretion, 2:66
- Endocrine disorders, 2:129–130
- Endocrine glands, function, 2:160
- Endocrine system
 chemicals that disrupt, 2:164
 function, 2:159–160
 location of hormones, 2:160
 metabolic diseases, 3:42
 multiple endocrine neoplasia, 4:153
- Endocytosis, defined, 4:166
- Endogenous, defined, 1:242, 4:41
- Endometriosis, defined, 4:19
- Endometrium, defined, 1:169
- Endonuclease enzymes
 apurinic/aprimidinic, 1:244
 defined, 3:47
 DNA repair, 3:114
 function, 3:112, 3:113
 structure preferences, 3:112–113
 See also Restriction enzymes
- Endophenotypes
 defined, 4:2
 example, 4:2
- Endoplasmic reticulum (ER), 1:105
 defined, 1:117, 3:202, 4:10
 structure and function, 1:112–113, 3:120, 3:121, 3:125, 3:202, 4:43–44
- Endoribonuclease enzymes, 4:57–58
- Endoscopic, defined, 1:168
- Endospores, of bacteria, 2:12, 2:14
- Endosymbionts
 chloroplasts as, 1:112, 2:12, 2:116, 2:194–196, 3:55, 3:56–57
 defined, 2:86
 intracellular, inheritance of, 2:198
 mitochondria as, 1:112, 2:12, 2:86, 2:116, 2:194–196, 3:52, 3:55, 3:56–57
 serial endosymbiotic theory, 3:56–57
- Endotoxins, eubacteria, 2:12
- Enhancer DNA sequences
 of antibody genes, 2:182
 function, 2:54, 4:114
 gene expression role, 2:64
 impact on gene therapy, 2:77
 mutations in, 2:27, 2:30, 2:54
- Enterobacteriaceae, 2:10
- Environmental activists, damage caused by, 4:132
- Environmental Protection Agency (EPA)
 biotechnology regulations, 1:11
 oil spill cleanups, 1:59
- Environment-gene interactions. *See* Gene-environment interactions
- Enzyme replacement therapy
 Gaucher disease, 3:45
 metabolic diseases, 3:39
 mucopolysaccharidosis, 3:45
 SCID, 4:75, 4:77
- Enzyme testing, Tay-Sachs disease, 4:101–102
- Enzymes
 to activate procarcinogens, 1:99
 combinatorial chemistry to synthesize, 1:175, 1:175
 defined, 1:45, 2:15, 3:2, 4:48
 to detoxify poisons, 1:45–46
 evolutionary origin, 2:22–25
 extremozymes, 1:38, 1:199
 genes as, 3:102–103
 industrial, genetically engineered, 1:63–64, 1:154, 4:134–135
 metabolism and, 3:37–38
 mutations, and metabolic disorders, 3:38
 one gene-one enzyme model, 3:76, 3:103
 proteins as, 2:52
 ribosomes as, 4:44–46
 ribozymes as, 2:25
 SDS-PAGE to study, 2:47–48
 See also Proteins; *specific enzymes*
- EPA (Environmental Protection Agency)
 biotechnology regulations, 1:11
 oil spill cleanups, 1:59
- Epidemiology/epidemiologists, 2:6–7
 defined, 1:101, 2:56, 3:217, 4:94
 methods used, 2:56–57
 public health role, 3:217–218
 statistical geneticist role, 4:94–95
- Epidermal growth factor, 2:66
- Epididymis, defined, 4:21
- Epigenetic modifications, X-chromosome inactivation, 4:176
- Epilepsy, disease symptoms, 2:168
- Epinephrine, function, 2:160
- Epistasis, 2:7–9
 Bombay phenotype, 2:8, 2:9
 as complex trait, 1:178–179, 1:180, 2:9
 defined, 2:184, 4:3
 distinguished from dominance, 2:7
 DNA methylation as, 3:46
 interactions among proteins, 2:9, 2:207
 mouse color coat, 2:8, 2:8
 nonpenetrance, 2:8
 quantitative traits, 4:1–3

- Epithelial cells
 abnormalities, cystic fibrosis, 1:**199–203**, 2:127–128
 carcinomas, 1:93
 colon cancer, 1:167, 1:169
 defined, 1:199
 telomerase production, 4:106
- EPP (erythropoietic protoporphyria), 3:38
- Epstein-Barr virus, and cancer, 1:92
- Equal Employment Opportunity Commission, genetic discrimination policies, 2:93
- Equine encephalitis virus, characteristics, 4:165
- ER. *See* Endoplasmic reticulum
- Ereky, Karl, 1:70
- ERISA (Employee Retirement Income Security Act), 2:93–94
- Erwinia*, to synthesize vitamin C, 1:63
- Erythema infectiosum (fifth disease), 4:165
- Erythrocytes
 damaged by sickle-cell disease, 2:137, 2:138
 hemoglobin binding, 2:136
- Erythromycin, ribosomal targets, 2:16
- Erythropoietic protoporphyria (EPP), 3:38
- Erythropoietin
 cost and access concerns, 1:69
 to treat sickle-cell disease, 2:138
- ES. *See* Stem cells, embryonic
- Escherichia coli*, 2:**9–11**
 as BAC vectors, 1:222–224, 1:223
 conjugation, 1:183, 1:183, 1:185, 3:151
 defined, 1:153, 2:86
 habitats, 2:9–10
 as model organism, 2:10, 2:10–11, 3:60
 as pathogen, 2:10
 varieties (strains), 2:10
 vitamin synthesis by, 2:10, 2:13
See also Eubacteria
- Escherichia coli*, genomes
 amount of DNA, 1:132
 binding protein tetramers, 3:202
 characteristics, 2:11, 2:15, 2:86, 2:116, 2:121, 2:172
 Chi sequences, 2:117
 chromosomal domains, 1:140–141
 chromosome shape, 1:139–140
 chromosome size, 2:116
 chromosome *vs.* cell size, 1:142
 DNA polymerases, 1:231–232, 3:113–114, 3:157
 DNA sequencing research, 2:10
 DnaK chaperones, 1:117
 endonuclease enzymes, 3:114
 gene cloning role, 1:153, 2:10, 4:129
 gene number, 2:116
 GroEl chaperones, 1:117
 Lambda phages, 4:119
 mutation rates, 3:99
 operons, 1:142, 3:131–135
 plasmids, 1:140, 3:151
 as recombinant DNA tool, 4:6
 repeated extragenic palindromic sequences, 2:117
 replication, 1:141, 2:10, 2:13
 sequencing, 1:140, 1:142, 2:11, 2:86, 2:116, 2:172
 transposons, 4:144–145
 uid A markers, 3:18
- ESI (electrospray ionization), 3:20
- Eskimos, inbreeding coefficient, 2:190
- Esophagus, cancer, 1:93
- Essentialist (typological) view of evolution, 4:93
- EST (expressed sequence tag) analysis, 1:223, 1:224, 1:229
- Estradiol
 17- β , function, 2:162
 and breast cancer, 1:90
- Estriol, unconjugated, 3:185
- Estrogen
 assisted reproduction and, 4:21
 and cancer risk, 1:180
 influence on gene expression, 2:161–163
 ovaries and, 2:160
 production in normal humans, 1:21, 1:22, 1:23
 replacement therapy, 1:90
- Estrogen receptors
 activity, diet and, 2:164
 function, 2:162
- Estrone, and breast cancer, 1:90
- ETC. *See* Electron transport chain
- Ethical issues
 Human Genome Project, 2:174, 2:177
 organ transplants, 4:141, 4:142–143
 population screening, 3:177–180
 preimplantation genetic diagnosis, 3:186–187
See also Biotechnology; Cloning; Gene therapy; Genetic discrimination; Genetic testing; Reproductive technology
- Ethicists, defined, 2:82
- Ethics, Natural law, 4:28
- Ethidium bromide, as mutagen, 3:88
- Ethnic differences. *See* Racial and ethnic differences
- Ethyl methanesulfonate (EMS), 1:19
- Ethylenediaminetetraacetic acid (EDTA), 3:221
- O⁶-ethylguanine, 1:19, 1:19
- Etiology
 defined, 1:40, 3:213
See also specific diseases and disorders
- Eubacteria, 2:**11–16**
 antibiotic resistance, 1:**26–29**, 1:140, 1:185, 4:44, 4:68–69
 as biopesticides, 1:57
 as bioremediators, 1:59, 1:61–62
 as chloroplast ancestor, 2:12, 2:195, 3:55, 3:56–57
 defined, 4:112
 distinguished from Archaea, 1:36–37, 2:11, 2:16
 distinguished from eukaryotes, 1:108, 2:11
 DNA profiling, 1:233, 1:239
 escaped transgene role, 1:68
 as intracellular symbionts, 2:198
 as mitochondrial ancestor, 2:12, 2:195, 3:52, 3:55, 3:56–57
 phylogeny, 1:36, 1:37
 recombinant, as biohazards, 3:152
 to synthesize amino acids, 1:63
 to synthesize antibiotics, 1:63
 as teratogens, 1:79–80
 vitamins synthesized by, 1:63, 2:13
See also Prokaryotes; *specific bacteria*; Transgenic microorganisms
- Eubacteria, genomes
 artificial chromosomes (BACs), 1:144–146, 1:222–224
 C value paradox, 2:114
 cDNA, 4:40
 characteristics, 1:139–144, 2:116, 3:132, 4:12, 4:128
 chromosomes, 2:15
 DNA, 2:14–15
 DNA methylation in, 3:47
 exceptions to universal code, 2:87
 gene expression, 2:15–16
 histone proteins lacking, 1:141, 2:15
 introns lacking, 2:31, 4:128
 mapping chromosomes, 4:119
 mutation rates, 3:99
 operons, 1:142, 2:15, 3:105, 3:**131–135**

- promoter sequences, 4:107
 ribonuclease role, 3:114
 RNA polymerase, 4:55
 RNase P, 4:45, 4:46, 4:51
 terminator sequences, 4:110
 transcription elongation rates, 4:56
 transcription factors, 4:112
 translation initiation sequences, 4:138
 transposable genetic elements, 3:22, 4:144–145
See also Plasmids; Restriction enzymes
- Eubacteria, metabolism**
 aerobic and anaerobic, 2:13
 chemolithotrophs, 2:13
 electron-accepting compounds, 2:13
 heterotrophs, 2:13
 phototrophs, 2:13
- Eubacteria, replication**
 binary fission, 2:13
 cell division, 1:113
 translation, 2:16
See also Conjugation; DNA replication, prokaryotes; Transduction; Transformation
- Eubacteria, structure**
 cell walls, 2:11
 cytoplasm, 2:12
 flagella, 2:12
 gram negative, 1:27, 2:12
 gram positive, 2:12, 2:13
 LPS layer, 2:11–12
 mitochondria, 2:12
 nucleoids, 2:14–15
 plasma membranes, 2:12
 ribosomes, 2:16, 4:42–44
 shapes, 2:13, 2:14
- Euchromatin**
 gene expression role, 2:63
 heterochromatin-euchromatin conversion, 1:135, 2:65
 structure and function, 3:121–122
- Eugenics, 2:16–21**
 as artificial selection, 4:69
 current status, 2:20–21
 defined, 2:90, 3:81, 4:26
 evolutionary fitness ranking, 2:17, 2:19
 eye color and, 2:32
 Fitter Family contests, 2:19
 gene counseling to prevent abuses, 2:90
 gene therapy and, 2:82–83
 germinal choice, 3:81
 Mendelian genetics and, 2:16, 2:18
 Muller's contributions, 3:81
 by Nazis, 2:16, 2:19–20, 2:90, 4:26
 negative, 2:17, 2:20
 origin of term, 2:16
 policies and laws, 2:16–19
 population screening and, 3:177–178
 positive, 2:17, 2:20–21, 3:81
 prenatal genetic testing, 3:186–187
 reproductive technology concerns, 4:26
 Sir Francis Galton and, 2:16–18
- Eukaryotes**
 defined, 1:152, 2:27, 3:16, 4:12
 distinguished from eubacteria, 2:16
 distinguished from prokaryotes, 1:108, 2:11, 2:12
 origin of term, 1:139
 phylogeny, 1:36, 1:37
 ribosomes, 4:42–43
 ribozymes, 4:44
See also Cells, eukaryotic
- Eukaryotes, genomes**
 of advanced groups, 2:114–115
 characteristics, 2:112–113, 2:121, 4:12
 distinguished from mitochondria and chloroplasts, 2:195–196
 gene expression, 2:53, 2:54
 gene structure, 2:63–64, 2:64
 haploid complements, 2:113
 promoter sequences, 4:107, 4:112
 protein length, average, 2:52
 repeated sequences, 2:115, 4:10–11
 RNA polymerases, 4:55–56
 RNA processing, 4:50, 4:57, 4:58
 RNA synthesis, 4:50
 terminator sequences, 4:110
 transcription factors, 4:112
See also Chromosomes, eukaryotic; DNA replication, eukaryotes
- European (predestined) model of development, 1:205–206**
- Euthanasia, eugenics and, 2:16, 2:18, 2:20**
- Evans, Martin, 4:126**
- Evolution**
 Darwinian, 1:131, 2:17, 2:26, 2:156, 3:32, 4:67, 4:92
 multiregional, 3:67
 typological (essentialist view), 4:92
 vertical, 4:92
See also Natural selection
- Evolution, human. *See* Anthropology, molecular**
- Evolution, molecular, 2:21–26**
 future goals, 2:25
 oldest known life, 2:21–22
 RNA as enzyme, 2:25
 RNA nucleotide origins, 2:22–23
 RNA replication without enzymes, 2:23–25
 subunit linking, 2:23
- Evolution, of chloroplasts**
 endosymbiont ancestors, 1:112, 2:12, 2:194–196, 3:56–57
 genome characteristics, 3:55
 genome size reduction, 2:116
See also DNA, chloroplast (cpDNA)
- Evolution, of genes, 2:26–31**
 accumulation of noncoding DNA, 2:114–115
 Alu sequences, 3:211–212, 4:11, 4:145–146
 balanced polymorphism, 1:45–46
 bioinformatics tools, 1:52, 1:56
 chromosomal aberration role, 2:28, 3:97
 crossing over, 2:28
 cytochrome C, 3:94
 DNA polymerases, 1:232
 and DNA repair, 1:248
 early studies, 3:32
 exon shuffling, 2:29, 2:30–31
 extranuclear genes to study, 2:198
 gene duplication, 2:28–29, 2:68–69
 gene families, 2:29, 2:30, 2:67–69
 genetic code, 2:86–87
 genomics tools to study, 2:121
 globin proteins, 2:68, 2:69, 2:136, 3:212–213, 3:212
 histone proteins, 3:94, 3:96
 homologies as indicators, 2:156–158
 intragenic distances, 2:115
 molecular clocks, 3:63, 3:67, 3:98
 mouse *vs.* human chromosomes, 3:96, 3:97
 natural selection and, 2:26, 3:93–94
 number of genes, 2:115
 overlapping genes, 3:136
 point mutations, existing genes, 2:26–27

- Evolution, of genes (continued)
 pseudogenes, 2:28, 2:29–30, 3:211–213
 quantitative traits, 4:1
 repeated sequences, 2:28, 2:30, 2:31, 2:115, 4:9, 4:10–11
 restriction enzymes, 4:31
 retroposition, 2:28
 retrovirus-like elements as indicators, 4:41
 ribosomal RNA as indicator, 1:36
 ribozyme role, 4:46
 sex chromosomes, 4:174, 4:177–179, 4:178
 species designations, 1:188, 1:189
 transcription factors, 4:112
 viroids, 4:163
- Evolution, of mitochondria
 endosymbiont ancestors, 1:112, 2:12, 2:194–196, 3:52, 3:55, 3:56–57
 genome characteristics, 2:116
See also DNA, mitochondrial (mtDNA)
- Ex vivo*, defined, 2:76
- Ex vivo* gene transfer, as gene therapy tool, 2:76
- Excelesis, functional genomics role, 2:124
- Exercise
 cancer and, 1:91
 cardiovascular disease and, 1:101
 diabetes and, 1:210
 familial hypercholesterolemia and, 2:57
 impact on gene expression, 1:225
 longevity and, 1:6
 sickle-cell disease and, 2:138
- Exit (E) site, in translation, 4:138
- Exogenous, defined, 4:121
- Exon shuffling, gene evolution role, 2:29, 2:30–31
- Exon skipping, 4:59
- Exons, 1:12
 in CFTR gene, 2:60
 defined, 2:30
 function, 1:12–13, 2:52, 2:53, 3:106, 3:181, 4:45, 4:58
 triplet repeats, 4:149, 4:150
- Exonuclease enzymes
 defined, 1:231
 function, 3:112, 3:113–114, 3:113
 proofreading, 1:231, 1:232, 3:113–114, 4:17, 4:18
 structure preferences, 3:112–113
- Exoribonuclease enzymes, 4:57–58
- Exploratory data analysis, 4:95–96
- Exports, 3:126
- Expressed sequence tag (EST) analysis, 1:223, 1:224, 1:229
- Expression analysis
 defined, 1:227
 DNA microarray tools, 1:228, 4:40
- Expression cloning, 1:157, 3:152
- Expression libraries, 1:224
- Expression vectors, 1:157
- Exteins, 3:179, 3:181
- Extinct in the Wild population status, 1:186
- Extinct population status, 1:186
- Extinct species
 cloning, 4:25
 DNA analysis, 3:68–69
 population bottlenecks, 3:167
- Extinction vortex, 1:188
- Extranuclear inheritance. *See* Inheritance, extranuclear
- Extremophiles. *See* Archaea
- Extremozymes, 1:38, 1:199, 3:157
- Exxon Valdez oil spill, 1:59, 1:60
- Eye color, 2:31–33
 aging and, 2:33
 change in response to disease, 2:33
 genetic components, 2:32–33
 impact of drugs, 2:33
 iris structure, 2:32
- Eye disorders
 birth defects, 1:75
 color blindness, 2:203, 3:78
 mitochondrial diseases, 3:54
 spondyloepiphyseal dysplasias, 2:131
See also Blindness; Retinoblastoma
- Eyes
 iris, 2:31–33
 retinas, 1:170, 4:98, 4:100
See also Vision; Vision, color
- F**
- Fabricated Man* (P. Ramsey), 4:26
- Fabry disease, symptoms and treatment, 3:41
- Factor VIII, hemophilia and, 4:174
- Factor-dependent terminators, 4:110
- Fallopian tubes
 defined, 4:19
 development, 1:22, 4:79
 function, 2:35
- Familial adenomatous polyposis (FAP), 1:167–169, 2:167–169, 4:153
- Familial aggregation studies, 1:213–214, 2:58, 2:60
- Familial hypercholesterolemia (FHC)
 as autosomal dominant disorder, 2:201
 clinical features, 2:202
 gene-environment interactions, 2:57
- Familial partial lipodystrophic diabetes (FPLD), 1:211
- Family studies
 CEPH, and gene mapping, 3:12, 3:13
 disease concordance studies, 1:213–214, 2:168–169, 3:7–8, 3:177
 epistatic interactions, 2:9
 eugenics applications, 2:17, 2:20
 genetic counseling role, 2:88–90
 genetic testing, 2:59, 2:99–100, 2:101, 2:168–169
 individual genetic variations, 2:192
 informed consent, 2:168
The Kallikak Family, 2:18, 2:19
 Mennonites, inbreeding, 2:190–191
 in molecular anthropology, 3:65
 pedigrees, 3:7, 3:36, 3:138–142, 3:141, 4:178
 positional cloning studies, 2:59–60
 sibling recurrence ratios, 2:58
 of Thomas Jefferson, 3:65
See also Adoption studies; Twin studies
- Family studies, specific diseases and traits
 addiction, 1:4–5
 ADHD, 1:40
 Alzheimer's disease, 1:16–18
 behavior, 1:47–48
 breast cancer, 1:91, 2:100, 2:102
 cardiovascular disease, 1:101–103
 colon cancer, 1:168–170
 cystic fibrosis, 1:202–203
 intelligence, 2:209–210
 longevity, 1:6–7
 psychiatric disorders, 3:213, 3:214, 3:214–215
 sexual orientation, 4:83–84
- FAP (familial adenomatous polyposis), 1:167–169, 2:167–169, 4:153
- Farnsworth-Munsell color vision test, 1:172
- Farnsworth's Panel D-15 color vision test, 1:172
- FAS/FAE (fetal alcohol syndrome/effects), 1:75, 1:81
- Fatal familial insomnia, 3:190

- Fats, breakdown in mitochondria, 1:111, 3:41
- Fatty acid metabolism
disorders, 3:40, 3:43
normal function, 3:41
- FDA. *See* Food and Drug Administration
- Federal Bureau of Investigation (FBI), CODIS DNA profiling, 1:235, 1:236, 1:237
- Feeder cells, in stem cell research, 2:3
- Fen1 protein, 4:17
- Fermentation
defined, 1:66
history, 1:66, 1:70, 1:71
- Ferns
alternation of generations, 2:115
polyploidy, 2:114, 3:164
- Ferris, James, 2:23
- Ferritin
function, 3:199
pseudogenes for, 2:30, 3:213
- Fertility. *See* Infertility; Pregnancy; Reproductive technology
- Fertility factors (conjugative plasmids), 1:140–142, 1:182–185, 1:184, 2:117, 3:151
- Fertilization, 2:33–36
donor insemination, 4:24
egg donation, 4:24
ejaculation, 2:35
external, in marine animals, 2:36
fraternal and identical twins, 2:35–36, 4:155–156
gamete fusion, 2:34, 2:35, 2:115, 2:133, 3:24
gene flow and, 2:70–71
haplodiploidy, 2:113
imprinting during, 2:184
intrauterine insemination, 4:24
microassisted, 4:22
normal human, 4:19
ovulation and, 2:34–35, 4:19, 4:21
“personhood” concept, 4:27–28, 4:29
polyploidy and, 2:69, 3:163, 3:164, 3:166
surrogacy, 4:24
See also Gametes; In vitro fertilization
- Fertilizers (plant nutrients), for transgenic plants, 2:106
- Fetal alcohol syndrome/effects (FAS/FAE), 1:75, 1:81
- Fetal cell sorting, 3:184–185
- FHC (familial hypercholesterolemia) as autosomal dominant disorder, 2:201
- clinical features, 2:202
gene-environment interactions, 2:57
- Fibrillin 1 gene (*FBR1*), and Marfan syndrome, 1:76
- Fibrin, role in blood clotting, 1:72
- Fibroblast growth factor receptors 3 (FGFR3), 1:75–76, 2:130
mutations, 2:67
- Fibroblast growth factors, function, 2:66
- Fibroblasts
defined, 2:65, 4:105
MyoD protein, 2:65
telomere shortening, 4:105
- Fifth disease (erythema infectiosum), 4:165
- Finland, founder effect, 2:37, 2:38
- FinnTwin studies, 1:50–51
- Fire blight, resistance in apples, 2:107
- Firefly protein, 1:69, 4:130
- FISH technique. *See* Fluorescence *in situ* hybridization
- Fisher, Ronald, 4:1, 4:95
- Fitness, defined, 2:146
- Fixed alleles, 3:171
- Flagella, eubacteria, 2:12
- Fluorescence *in situ* hybridization (FISH), 1:128–129, 1:145, 2:98, 2:187, 2:188, 3:182–183
- Fluorescence-activated cell sorters, 3:184
- Fluorescent DNA sequencing, 4:72–73, 4:73
- Fluorophore, defined, 2:149
- Fly lab, 3:73, 3:80
- Flynn effect, 2:209
- FMRP, fragile X syndrome and, 2:42
- FMR-1* gene, and fragile X syndrome, 1:76, 2:41–42, 3:49, 4:150
- Folate
and birth defects, 1:78
to synchronize cells, 1:128
- Follicle-stimulating hormones, 4:19, 4:21–22
- Food and Drug Administration (FDA)
Ames test requirements, 1:21
biotechnology regulations, 1:11, 1:72, 4:130–131
- Food safety
antibiotic resistance concerns, 1:28
genetically modified foods, 1:10–11, 1:66–68, 1:67, 1:73
- Ford, C. E., 1:119, 4:80
- Forensics
defined, 3:220
DNA purification tools, 3:220–221
genetics tools, 1:42
See also DNA profiling (fingerprinting)
- Formaldehyde, and ribose synthesis, 2:23
- Fossils, stromatolites, 2:21–22, 2:22
- Founder effect, 2:36–39
Amish populations, 2:37, 3:174
Ashkenazic Jews, 2:37, 3:174, 4:101
clinical and genealogical record-keeping, 2:38
deafness, Martha’s Vineyard, 2:37
early human populations, 3:67, 3:168–169, 3:169
gene flow, 2:70–71
genotype frequencies and, 3:174
haplotype analysis, 2:38
Hutterite colonies, 2:37, 2:37, 2:38–39
inbreeding, 2:189–191
Lac Saint-Jean population, 4:101
linkage disequilibrium mapping, 2:38
mitochondrial Eve, 3:68
Tristan da Cunha islanders, 2:190, 3:174
- Fragile XE syndrome, 4:150
- Fragile X syndrome, 1:76, 2:39–42
biology of disorder, 2:42, 4:150
discovery of, 2:39–40
DNA methylation role, 3:49
genomic screening, 2:168
inheritance patterns, 2:40–42, 4:175
Sherman Paradox, 2:40, 2:40
symptoms, 2:39, 2:41, 2:211, 4:150
triplet repeat component, 2:41–42, 4:11, 4:175
- Fragmentation of habitat/populations, 1:186–187, 1:188
- Frameshift mutations
Charcot-Marie-Tooth disease, 4:4
described, 2:86, 3:95, 3:96
Gaucher disease, 4:4
hypercholesterolemia, 4:4–5
mutagen role, 3:88
neurofibromatosis, 3:153
phenylketonuria, 4:5
in pseudogenes, 3:209
Tay-Sachs disease, 4:101
terminology, 4:5

- Franklin, Rosalind, 4:172
- Free radicals, as mutagens, 3:88, 3:89
- Friedreich's ataxia
clinical features, 2:202
and diabetes, 1:212
as triplet repeat disease, 4:150
- Frog, African clawed (*Xenopus laevis*), 3:60, 3:61
- Frontal lobe, defined, 1:40
- Frost tolerance, transgenic plants, 2:106, 2:107
- Fruit flies (*Drosophila*), **2:42–45**
addiction studies, 1:5–6
aging studies, 1:7
developmental processes, 1:204–208, 1:207, 1:208, 2:44–45, 2:65–66
early studies, 2:42–43
as model organism, 2:43, 2:44, 3:60, 3:73
sex determination, 4:81–82
- Fruit flies, genomes
characteristics, 2:43, 2:121, 2:122
DNA methylation, 3:46
effects of radiation on, 2:173, 3:81, 3:90–92, 3:90, 3:166–167
exceptions to universal code, 2:87
Fly lab, 3:73, 3:80
gene identification studies, 1:155
genes, number, 2:115, 2:174
homeotic genes, 2:45, 2:65–66, 2:68
mapping studies, 2:43, 3:7, 3:12–15, 3:74, 3:75, 3:80, 3:83
naming genes, 3:107
nondisjunction studies, 1:131
number of genes, 1:11–12
phenotype manifestations, 2:43
polyploidy, 3:166
polytene chromosomes, 1:126, 1:136–137, 1:136, 2:43–44
restriction maps, 1:155–156, 1:156
RNA interference, 4:53
sequencing, 2:172
transposons, 2:43, 3:97–98, 4:145, 4:146, 4:148
X-linked inheritance studies, 1:130–131, 2:43, 3:73–74, 3:80, 3:91–92, 3:102
- FSH hormones, and endocrine disorders, 2:129
- FSIQ (full scale) intelligence scores, 2:208
- Fuchs' heterochromic iridocyclitis, eye color changes and, 2:33
- Fucosyltransferase, 2:207
- Functional group substitution, combinatorial chemistry, 1:173–174, 1:174
- Fungi
alternation of generations, 2:115
as biopesticides, 1:57
heterokaryosis, 2:115
inheritance of mitochondrial and chloroplast genes, 2:197
as opportunistic pathogens, 2:155
plasmids in, 3:150
- FX174 virus, genome characteristics, 2:117
- G**
- G protein coupled receptor (GPCR) pathway, signal transduction role, 4:88–89, 4:88, 4:90
- G proteins, signal transduction role, 4:88–89
- G2/S boundary, 4:154
- G6PD (glucose-6-phosphate dehydrogenase deficiency), heterozygote advantage, 2:148
- G418, as gene targeting tool, 4:125
- Gag* gene, 4:35, 4:36, 4:144, 4:145
- Gain-of-function mutations
defined, 2:74
inheritance patterns, 3:96
in pseudogenes, 3:211
sickle-cell disease, 2:78–79, 2:80
toxic, 2:201
transposable genetic elements, 4:144
- Gajdusek, D. Carleton, 3:189
- Galactose operons, 3:134
- Galactosemia, symptoms and treatment, 3:40, 3:44
- Galactose-1-phosphate uridyl transferase, 3:44
- β -galactosidase enzyme, 3:132, 3:134, 3:135
- Gall, Joseph, 1:135
- Gallo, Robert, 4:40
- Galton, Francis
eugenics movement, 2:16–18
gemmule experiments, 1:129–130
statistics contributions, 4:95
twin studies, 2:207–208, 2:210
- Gamete intrafallopian transfer (GIFT), 4:23
- Gametes
chromosomal makeup, 2:33–34
crossing over and, **1:194–197**
defined, 1:120, 2:33, 3:24, 4:78
fertilization role, 3:24
fusion of, 2:34, 2:35, 2:133
gene flow between, 2:70–71
haploid complements, 2:113, 2:115, 3:24, 3:60, 3:163
imprinting of, 2:184, 3:48
independent assortment of, 3:34–35
mutations, consequences, 3:93
mutations, de novo, 1:120
segregation of, 3:34
unbalanced, 1:123
See also Eggs; Fertilization; Meiosis; Sperm
- Gametogenesis, aneuploidy, 1:121
- Gametophytes, defined, 2:115
- Gamma rays, as mutagens, 3:88
- Gamma-retrovirus, characteristics, 4:35
- Gamow, George, 1:193
- Ganciclovir, 2:79, 3:17, 4:126
- Gangliosides, and Tay-Sachs disease, 3:45, 4:98–100
- Gangrene, causal organism, 2:13
- Garrod, Sir Archibald, 3:37, 3:43
- Gastro-intestinal disorders
cystic fibrosis, 1:200
mitochondrial diseases, 3:54
stomach cancer, 169, 1:93
See also Cancer, colon
- Gastrulation, defined, 4:182
- Gating process, of nuclear pores, 3:126
- Gaucher disease
as frameshift mutation, 4:4
symptoms and treatment, 2:202, 3:40, 3:45
- Gaurs, cloning, 1:164
- Gautier, Marthe, 1:119
- G-banding of chromosomes, 1:127–128, 1:127, 1:135
- GCN5-regulated genes, 1:138
- GCP* (green cone pigment) genes, 1:171
- GDP and GTP (guanine diphosphate and triphosphate), signal transduction role, 4:88, 4:89–90
- Gel electrophoresis, **2:45–50**
acrylamide gels, 1:86, 2:47
agarose gels, 1:86, 2:49
anthrax analysis, 3:71
for automated sequencers, **1:43–45**, 1:53, 1:198
for blotting procedures, **1:86–89**, 1:152
considerations in choosing, 2:46
defined, 1:43, 3:18, 4:33
in DNA purification, 3:221
in DNA sequencing, 4:70–72
electrophoresis, defined, 1:198
ethidium bromide, 3:88

- to identify polymorphisms, 3:160–162, 3:161
- isoelectric focusing, 2:48, 3:208
- nucleic acid separation, 2:49
- PAGE, 2:46–47
- PAGE, 2-D, 2:48–49, 2:48, 3:207–208
- PAGE, SDS, 2:47–48
- in PCRs, 3:158
- procedure, 2:45–46, 2:46, 2:47
- as proteomics tool, 3:18–19, 3:206
- pulse-field, 2:49–50, 4:33
- in STR analysis, 1:236
- Gelsinger, Jesse, 1:69, 2:77
- Gemmules, 1:129–130
- Gems, subnuclear bodies, 3:124
- Genazyme, 1:29
- GenBank database, 1:52, 1:54, 2:156, 2:174
- Gender differences
- color vision defects, 1:172
 - crossing-over rates, 1:197
 - intelligence, 2:209
 - recombination frequencies, 3:7
 - See also* Sexual development; X chromosomes; Y chromosomes
- Gender identity. *See* Sexual orientation
- Gene annotation, defined, 2:116
- Gene capture, 3:127
- Gene cloning. *See* Cloning genes
- Gene clusters. *See* Gene families
- Gene discovery, 2:57–61
- association studies, 2:61
 - candidate gene approach, 2:59, 2:60
 - defining traits, 2:58
 - genomic screen approach, 2:59, 2:61
 - genomics industry role, 2:123–125
 - linkage and recombination tools, 3:4–8
 - mapping tools, 3:11
 - naming genes, 3:107–108
 - ortholog tools, 2:158
 - polymorphisms as tools, 3:162
 - positional cloning tools, 2:59–60
 - reverse transcriptase as tool, 4:40
 - See also* Human disease genes, identification of; Human Genome Project; *specific diseases and traits*
- Gene dosage
- nondisjunction and, 3:111
 - of polyploids, 3:164
- Gene duplication
- Bar eyes mutation, 3:81
 - consequences, 3:111, 4:67–68
 - gene evolution role, 2:28–29, 2:28, 2:68–69
 - globin gene family, 3:212
 - pseudogenes, 3:209–210, 3:211
 - tandem arrays (gene clusters), 2:28, 2:68–69, 2:115, 2:132, 2:192
- Gene expression
- by bacteriophages, 4:118–120
 - and cancer growth, 1:95
 - defined, 1:56, 2:61, 4:54
 - ectopic, 2:74
 - gene family differences, 2:68–69
 - gene-environment interactions, 1:160
 - longevity, gene therapy, 2:77, 2:79, 2:81
 - in marker systems, 3:15–18
 - vs.* mRNA abundance, 3:206
 - patterns, progeria, 1:3
 - point mutation role, 2:27
 - of retroviruses, 4:36
 - transformation competence, 4:121
 - in transgenic plants, 2:108, 4:133
- Gene expression, control overview, 2:61–67
- during development, 2:65–66
 - DNA coiling in nucleus, 3:122
 - early studies, 3:103–106
 - environmental factors, 2:61–62
 - eubacteria, 2:15–16
 - eukaryotes, 2:54
 - expression failures, consequences, 2:67
 - flow from DNA to proteins, 2:62, 2:62
 - growth factors, 2:66
 - hormonal regulation, 2:66–67, 2:161–164, 4:113–114
 - intein and extein role in, 3:181
 - mRNA role, 3:70
 - operons, 1:142, 2:15, 3:105, 3:131–135
 - prokaryotes, 2:15, 2:52–53, 2:54, 2:62
 - RNA interference, 1:31, 4:52–54
 - transcription repression/promotion, 4:110–111
 - See also* Epistasis; Imprinting; Mosaicism; Pleiotropy; Signal transduction; Splicing, alternative; Transcription factors; X chromosomes; inactivation of
- Gene expression, research tools
- artificial chromosomes, 1:145–146
 - autoradiography, 1:136–137
 - blotting, 1:87–89
 - DNA fingerprinting, 1:221
 - DNA libraries, 1:224
 - DNA microarrays, 1:225–229
 - E. coli*, 2:10
 - gene cloning tools, 1:157
- Gene families, 2:67–70
- defined, 1:132
 - evolution of, 2:29, 2:30, 3:212–213, 3:212
 - expressed and nonexpressed genes, 2:67, 2:68–69
 - globin proteins, 2:67, 2:68, 2:69, 3:211, 3:212–213, 3:212
 - location on chromosomes, 2:68
 - sizes and numbers, 2:67–68
- Gene flow, 2:70–71
- in fragmented plant/animal populations, 1:187–189
 - speciation and, 4:92–93
 - See also* Founder effect; Genetic drift; Hardy-Weinberg equilibrium
- Gene guns, 1:9, 1:58, 1:255, 2:107, 4:132–133
- Gene knock-ins. *See* Knock-in mutants
- Gene knock-outs. *See* Knock-out mutants
- Gene machines (oligonucleotide synthesizers), 3:155
- Gene mapping. *See* Mapping genes
- Gene mining, 2:123–124
- Gene nature, history, 3:101–106
- DNA as heredity material, 1:250–251, 1:250, 3:103, 3:104, 3:105, 4:120, 4:124
 - gene action and mutation, 3:102–103
 - gene coding studies, 1:252–253
 - genes as hereditary particles, 3:101–102
 - See also* Chromosomal theory of inheritance; DNA structure and function, history; Mendelian genetics
- Gene patents. *See* Patenting genes
- Gene pools, 3:171–172
- Gene silencing. *See* Imprinting; Methylation, of DNA; RNA interference; X chromosomes; inactivation of
- Gene splicing. *See* Splicing, alternative; Splicing, RNA
- Gene targeting
- adding/deleting genetic material, 2:73
 - homologous recombination method, 2:71–72, 2:72, 2:73, 3:91, 4:125–126

- Gene targeting (continued)
 positive and negative selection, 4:125–126
 procedures, 2:4
 RNA interference tools, 4:54
 rodent models, 4:61, 4:124–127
 therapeutic uses, 2:73
 vectors, 2:71, 2:72, 4:125–126
- Gene therapy, 2:74–80
 advantages, 2:83
 antisense nucleotide tools, 1:29, 1:30–31
 disadvantages, 2:77
 disease targets, 2:74
 ectopic expression toxicity, 2:74
 gene targeting tools, 2:73, 2:81
 genomics industry role, 2:124–125
 germ line *vs.* somatic cell therapy, 2:80–81, 2:82–83
 lipid vesicle vectors, 2:76–77
 longevity of gene expression, 2:77, 2:81
 patent issues, 3:137
 recombinant adenovirus vectors, 2:75–77, 2:78
 recombinant DNA tools, 4:5–6
 retrovirus vectors, 2:76
 ribozyme tools, 2:79, 4:46
 in utero, 2:81, 2:82
 See also Stem cells, embryonic
- Gene therapy, ethical issues, 2:80–83
 clinical trial safety, 2:81
 disease treatment *vs.* trait enhancement, 2:82–83
 embryonic stem cells, 2:6
 germ line *vs.* somatic cell therapy, 2:80–81, 2:82–83
 informed consent, 1:69, 2:81–82
 success concerns, 2:83
- Gene therapy, specific diseases
 cancer, 2:74, 2:79, 2:81
 cardiovascular disease, 1:103
 cystic fibrosis, 1:201, 2:81
 hemophilia, 2:78, 2:80, 2:145
 HIV/AIDS, 2:74, 2:80, 4:46
 metabolic diseases, 3:39
 muscular dystrophies, 2:75, 2:77–78, 2:80, 2:81, 3:86–87
 progeria, 1:3
 SCID, 2:76, 2:81, 4:75–78
 sickle-cell disease, 2:78–79, 2:80, 2:138
 Tay-Sachs disease, 4:102
 thalassemias, 2:140
- Gene trees, 4:178, 4:178
- Gene walking, 1:202
- Genecor, 1:73
- Gene-environment interactions, 2:54–57
 disease role, 1:213, 1:214
 and divergence from common ancestry, 3:63
 as eugenics rebuttal, 2:20, 3:81
 in gene discovery studies, 2:58
 and gene expression, 1:160, 2:61–62
 methods for identifying, 2:56
 and mutation rates, 3:99–100
 patterns, 2:56–57
 public health role, 3:217
 quantitative trait role, 4:1–3
 twins, 2:58, 4:157–158, 4:160, 4:161–162
 See also Complex (polygenic) traits
- Gene-environment interactions, specific diseases and traits
 alcoholism, 1:51
 Alzheimer's disease, 1:18, 1:181, 2:56
 birth defects, 1:77
 cancer, 1:90, 1:92, 1:179–180
 cardiovascular disease, 2:102
 diabetes, 2:102
 familial hypercholesterolemia, 2:57
 intelligence, 2:207, 2:209–210, 2:211
 phenylketonuria, 2:55–56
 psychiatric disorders, 3:214
 sexual orientation, 4:83–84
- Genentech, 1:72, 2:125, 4:129
- General cognitive ability (g), 2:207–211
 See also Intelligence
- General Electric Corporation
 oil spill cleanup, 1:61
 PCB pollution, 1:59
- Genes, 2:50–54
 additive effects, 1:178–179, 1:178
 as catalysts, 3:102, 3:103
 causative and susceptibility, 1:17, 1:18, 1:179, 2:55, 2:100, 2:118–120, 2:124
 for energy metabolism, 2:194
 as enzymes, 3:102–103
 length, 2:52
 minimal number required, 1:143
 modifier, 3:81
 multiplicative effects, 1:178–179, 1:179
 mutable, 3:21
 naming, 3:107–108
 one gene-one enzyme model, 3:76, 3:103
 one gene-one polypeptide model, 3:103
 one gene-one protein model, 3:103
 origin of term, 1:130, 2:50, 3:73, 3:101
 overlapping, 2:85–86, 3:135–136
 paralogous, 2:158, 3:210
 physical *vs.* genetic linkage, 3:5–6, 3:5
 similarities to proteins, 1:52–53
 size *vs.* mutation rate, 3:99–100
 structure and function, 2:50, 2:51–52, 2:52, 2:53, 3:70
 syntenic, 3:5
 See also Eukaryotes, genomes; Evolution, of genes; Genomes; Human genome; Prokaryotes, genomes; Pseudogenes; *specific organisms*
- Genesis 3000 sequencers, 1:44
- Gene-specific factors
 activators and repressors, 4:114
 function, 4:113
- Genetic Alliance, 2:105
- Genetic code, 2:83–87
 coding and noncoding regions, 2:50, 2:52, 2:114–115, 3:82–83, 3:93, 3:99, 3:105, 3:160
 codon table, 2:85, 4:137
 defined, 4:137
 degeneracy, 2:27, 2:84, 4:4, 4:5, 4:135
 DNA and RNA alphabets, 2:83–84, 4:135
 exceptions to universal code, 2:86–87
 gaps in, 2:85–86
 history, 1:252–253
 homologies, comparing, 2:157–158, 3:61
 human, sample, 2:113
 mutations, and evolution, 2:26–30, 2:27
 mutations, consequences, 3:94–95
 overlapping genes, 2:85–86, 3:135–136
 for protein, 2:50, 2:52
 punctuation lacking, 2:86
 for RNA molecules, 2:52
 sequence comparisons, mammals, 2:157
 translation by RNA, 2:84
 as universal, 1:61, 1:152, 2:21, 2:86, 3:60
 wobble hypothesis, 2:85, 4:4

- See also Codons; Reading frames; Transcription
- Genetic counseling, **2:87–91**
 chromosomal aberrations, 1:125
 clinical geneticist role, 1:149–151
 Human Genome Project and, 2:92
 Huntington's disease, 2:103–104
 need for, 2:88–89, 2:88, 2:189, 2:192
 pedigrees as tools, 3:139–140, 3:142
 prenatal testing, 2:97, 3:185, 3:186
 presymptomatic testing concerns, 2:100–101
- Genetic counseling, specific diseases
 AIS, 1:21, 1:26
 Alzheimer's disease, 1:18
 birth defects, 1:74
 cardiovascular disease, 1:102, 1:103
 fragile X syndrome, 1:76
 hemophilia, 2:143
- Genetic counselors, **2:91–92**
 code of ethics, 2:90
 roles, 2:89–90, 2:91–92, 3:186
 statistics, 2:89, 2:91
- Genetic determinism, 2:102
- Genetic discrimination, **2:92–94**, 2:102
 Alzheimer's disease, 1:18
 Americans with Disabilities Act (ADA), 2:93
 Employee Retirement Income Security Act, 2:93–94
 Equal Employment Opportunity Commission on, 2:93
 genetic counselor role, 2:92
 Health Insurance Portability and Accountability Act, 2:94, 3:191–192
 Human Genome Project concerns, 2:177
 legal issues, 3:4, 3:192, 3:193
 legislation to prevent, 1:68
 pharmacogenomics and, 3:146
 population screening and, 3:177–178
 potential solutions, 2:94
 as presymptomatic testing concern, 2:100
 public health role, 3:216
 Rehabilitation Act of 1973, 2:93
 See also Genetic testing
- Genetic diversity
 agricultural biotechnology and, 1:11
 biodiversity, defined, 1:11
 of chimpanzees, 3:167–168, 3:168
 conjugation role, 1:142
 crossing-over role, 1:194–196, 2:28, 2:28, 3:24, 3:25, 3:26, 3:29, 3:60
 duplicate genes and, 2:28–29
 evolution of genes and, 2:26, 2:31
 homologous structures, 2:156
 Human Genome Project concerns, 2:175
 of humans, 1:11–12, 3:167–170, 3:168, 3:174, 3:177–178
 immune system genes, 2:179–181
 inbreeding and, 2:190
 of maize, 3:9–10, 3:9
 natural selection and, 4:67
 polymorphisms and, 2:126
 polyploidy and, 2:69
 of threatened plant/animal populations, 1:186–189, 1:190–191
 vs. tolerance of new alleles, 2:28
 See also Founder effect; Individual genetic variation; Population bottlenecks
- Genetic drift, **2:94–96**
 allele frequency role, 2:95
 defined, 2:36, 4:92
 distinguished from natural selection, 2:70–71
 and divergence from common ancestry, 3:63, 3:174
 founder effect role, 2:36–37, 3:174
 Hardy-Weinberg equilibrium, 2:135, 2:136, 3:174
 random events and, 2:95–96, 3:174
 speciation and, 4:92–93
- Genetic engineering. See Biotechnology
- Genetic equivalence
 defined, 1:205
 and gene expression, 1:225
- Genetic exceptionalism, 2:102
- Genetic heterogeneity, defined, 2:61
- Genetic ID, genetic testing, 3:2
- Genetic load, calculated by Muller, 3:81
- Genetic markers
 behavior genetics, 1:48
 cardiovascular disease, 1:103
 CODIS DNA profiling, 1:235, 1:236, 1:237
 cystic fibrosis gene, 1:202
 linkage disequilibrium mapping, 2:38
 polymorphic, 3:6, 3:7
 polymorphic markers, 2:58, 2:59, 2:60, 4:178
 RFLPs, 3:6, 3:13–14, 3:161–162, 3:161, 4:34
 SNPs, 2:118–119, 3:6, 3:14, 3:65, 3:162
 STRPs, 3:6, 3:14
 VNTRs, 1:236, 3:6, 3:14, 3:162, 4:8–9
 See also Marker systems; Short tandem repeats (microsatellites)
- Genetic medicine. See Clinical geneticists
- Genetic predisposition
 causative and susceptibility genes, 1:17, 1:18, 2:55, 2:100, 2:118–120, 2:124
 complex traits, 1:177–181, 1:213–215
 defined, 1:92
 discrimination based on, 2:92–94, 2:177, 3:190, 3:192, 3:193
 DNA microarray tools, 1:228
 epistatic interactions, 2:9
 pharmacogenomics to predict, 3:144–147
 population screening, 1:202–203, 2:6–7, 2:99–100, 2:143, 3:175–178
 public health role, 3:216–217
 quantitative trait role, 4:1–3
 reduced penetrance, 2:61
- Genetic predisposition, to specific diseases and traits
 cancer, 1:90, 1:91–92, 1:97–98, 1:99
 cancer, colon, 1:99, 1:167–170
 cardiovascular disease, 1:101–103
 diabetes, 1:209–212
 Parkinson's disease, 4:161
- Genetic structure, in populations, 3:172
- Genetic testing, **2:96–101**
 carrier testing, 2:99–100
 clinical geneticist role, 1:149–151
 cytogenetics analysis, 1:119
 to detect polyploidy, 3:166
 genetic counselor role, 2:90
 genomic medicine, 2:118–120
 genomics industry role, 2:124
 Guthrie test, 3:176
 hemoglobin electrophoresis, 3:176
 Human Genome Project contributions, 3:176, 3:187, 3:215, 3:216
 laboratory technician role, 3:2

- Genetic testing (continued)
 mutation detection procedures, 3:92–93
 newborns, 2:98–99, 3:42–43, 3:44
 patent issues, 3:137
 PCR contamination concerns, 3:158
 presymptomatic, 2:100, 2:100–101
 presymptomatic, children, 2:100–101
 procedures, 2:102
 public health role, 3:218
 symptomatic, 2:99
See also Population screening; Prenatal diagnosis
- Genetic testing, ethical issues, **2:101–106**
 CLIA standards, 3:218
 clinical geneticist role, 1:149–150
 eugenics concerns, 3:177–178, 4:29–30
 genetic counselor role, 2:92, 2:102–104, 2:105
 genetic determinism, 2:102
 Human Genome Project concerns, 2:177
 informed consent, 2:102, 3:175
 personal autonomy, 2:101, 2:103
 population screening, 3:177–178
 prenatal and childhood testing, 2:100–101, 2:103
 privacy concerns, 1:68, 2:100, 2:177, **3:190–193**
 risks and benefits, 2:100, 2:102–103, 3:218
 support mechanisms, 2:105
 termination of pregnancies, 2:103, 2:177
 who should receive, 2:96
See also Genetic discrimination
- Genetic testing, specific diseases
 AIS, 1:21, 1:25
 Alzheimer's disease, 1:18
 behavior, 1:48
 blood type, 1:83
 cancer, breast, 1:91–92, 2:89–90, 2:100
 cancer, colon, 1:168–170
 cardiovascular disease, 1:102, 2:105
 chromosome disorders, 1:79
 congenital adrenal hyperplasia, 3:176, 3:177
 congenital hearing loss, 1:76
 cystic fibrosis, 1:202–203, 3:177
 diabetes, 1:210
 Down syndrome, 1:256, 1:257–258
 fragile X syndrome, 1:76, 2:40
 hemoglobinopathies, 3:177
 Huntington's disease, 2:100, 2:101, 2:103–104
 hypothyroidism, 3:176–177
 muscular dystrophies, 2:101, 3:85, 3:86
 phenylketonuria, 1:75, 2:55–56, 2:99, 2:119, 3:42–43, 3:176
 retinoblastomas, 2:100–101
 sickle-cell disease, 3:177
 Tay-Sachs disease, 2:96, 2:99–100, 3:174, 3:177, 4:98, 4:101–102
 triplet repeat diseases, 4:152
 X-linked disorders, 2:101
- Genetically modified (GM) foods, **2:106–110**
 benefits, 1:9–11, 1:73
 biopesticides, 1:57–58
 consumer acceptance of, 2:107, 2:108–109
 ethical issues, 1:11, 1:67–68, 1:67, 1:73, 3:3, 4:130–132
 field production, 2:108
 labeling, 1:67, 1:68, 2:109, 4:130
 modification technique, 2:107–108
 plant genetic engineer role, **3:149–150**
 recombinant DNA technology, 4:6–7
 regulations, 2:107, 2:108–109
 statistics, 2:108
 substantial equivalence principle, 4:130–131
 traits, 2:106–107
- Genetically modified organisms (GMOs). *See* Transgenic organisms
- Geneticists, **2:110–111**
 clinical, 1:74, **1:149–152**, 1:213–215, 2:90, 2:91–92, 3:193–194
 conservation, **1:190–192**
 Internet tools, 2:212
 statistical, 3:193, 3:194–195, **4:93–95**
See also specific geneticists
- Genetics, **2:111–112**
 biochemical, 3:103
 origin of term, 1:130
See also Classical hybrid genetics; Mendelian genetics; Population genetics
- Genetics professors. *See* College professors
- Genital and urinary tract disorders, birth defects, 1:75
- Genitals
 AIS individuals, 1:25
 normal human, 1:21, 1:22, 1:23, 4:78–79
- Genocide, eugenics and, 2:16
- Genograms, distinguished from pedigrees, 3:139
- Genome Web site, 1:142, 2:156, 2:176, 2:212
- Genomes, **2:112–117**
 chromosome number range, 2:113–114
 comparisons, molecular anthropology, 3:63–70
 databases of sequences, 1:142, 2:123–124, 2:156
 defense, DNA methylation role, 3:46–49
 defined, 1:9, 2:11, 3:22, 4:6
 distinguished from proteomes, 3:205
 homologies, comparing, 2:157–158
 McClintock's concept of, 3:22
 nuclear genes, gene density, and intergenic sequences, 2:114–115
 in nucleus, 1:112, 1:132–133, 3:119–126
 size range, 2:114
See also Chloroplasts, genome; Eukaryotes, genomes; Human genome; Mitochondrial genome; Ploidy; Prokaryotes, genomes; *specific organisms and organelles*
- Genomic clones, *vs.* cDNA clones, 1:154
- Genomic medicine, **2:118–120**
 DNA microarray tools, 2:119
 genetic microchips, 2:119–120, 2:119
 Human Genome Project contributions, 2:118, 2:119
 SNP role, 2:118–119, 2:122
See also Pharmacogenetics and pharmacogenomics
- Genomic parasites. *See* Transposable genetic elements
- Genomic plasticity, 2:117
- Genomic screening
 gene discovery role, 2:59, 2:169–170
 gene evolution role, 2:61
 selected disease-causing genes, 2:168
See also Genetic testing
- Genomica, bioinformatics, 2:124
- Genomics, **2:120–123**
 applications, 2:121–122, 4:133

- defined, 2:177, 3:143
functional, 2:122, 2:124
high-throughput screening tools, 2:120, 2:150, 3:205
model organism sequencing, 2:120
repeated sequences as tools, 4:11
See also Human Genome Project
- Genomics industry, 2:123–125**
bioinformatics companies, 2:124
defined, 2:123
established pharmaceutical companies, 2:124–125
functional genomics companies, 2:122, 2:124
future of, 2:125
gene mining companies, 2:123–124
large-scale sequencing companies, 2:123
new pharmaceutical companies, 2:125
population-based genomics companies, 2:124
- Genotype, phenotype and, 2:125–129**
adverse drug reactions, 3:144–145
CEPH families, 3:12
eugenics applications, 2:20
gene discovery, 2:57–61
induced mutagenesis to study, 3:90–93
in maize, 3:9–10
in Mendelian genetics, 3:33–37, 3:102
silent mutations, 2:127, 2:157, 3:93, 3:96, 3:99
See also Alleles; Dominant alleles; Individual genetic variation; Phenotypes; Polymorphisms; Recessive alleles
- Genotypes**
adaptation to environment, 4:67
of AIS individuals, 1:21, 1:23
blood groups, 1:83
defined, 1:21, 2:95, 3:140, 4:2
frequencies in gene pool, 3:171–172
genomic medicine applications, 2:118–120
Hardy-Weinberg predictions of, 2:135
represented in pedigrees, 3:140
standards for specifying, 2:133
variations, consequences, 2:126
- Genotypic means, 4:2**
- Genotyping techniques, 3:160–161**
- GenPharm, transgenic cows, 1:73**
- Genzyme Transgenics, biopharmaceuticals, 2:125**
- Germ cells**
defined, 2:80, 3:93, 4:106
mutations, consequences, 2:192, 3:93, 3:99
telomerase production, 4:106
- Germinal choice, 3:81**
- Germ-line gene therapy, advantages/disadvantages, 2:80–81**
- Gertsman-Sträussler-Scheinker (GSS) syndrome, 3:189–190**
- Gestational diabetes (GDM), 1:209, 1:212**
- GEY gene, for eye color, 2:33**
- GFP (green fluorescent protein), 2:73, 3:16, 3:17–18**
- GH (growth hormone), and endocrine disorders, 2:129–130**
- Giemsa stains, 1:127, 1:135**
- GIFT (gamete intrafallopian transfer), 4:23**
- Gilbert, Walter, 1:72, 2:172, 2:173, 4:65**
- Glaucoma, genomic screening, 2:168**
- Glaxo-SmithKline, drug development, 2:124**
- Globin proteins, 2:139**
alpha, beta, and gamma chains, 2:136, 2:148, 3:153, 3:212–213, 3:212
defined, 2:29
evolution of, 2:68, 2:69, 2:136, 3:211–213, 3:212
function, 2:136
gene family, 2:67–68, 3:211, 3:212
pseudogenes in, 3:211
sequence comparisons, mammals, 2:157
- Glucocorticoids, function, 2:160**
- Glucokinase, maturity onset diabetes and, 1:211**
- Glucose**
breakdown in mitochondria, 1:111–112
defined, 1:110
diabetes and, 1:209, 1:212
glucocorticoids to regulate, 2:160
- Glucose-6-phosphate dehydrogenase deficiency (G6PD), heterozygote advantage, 2:148**
- Glutamic acid (Glu)**
chemical properties, 3:200
genetic code, 2:85, 2:87, 4:137
substitution, sickle-cell disease, 2:137, 2:139, 2:148, 3:200
- Glyceraldehyde 3-phosphate dehydrogenase genes, 3:213**
- Glycine (Gln)**
chemical properties, 3:200
genetic code, 2:85, 4:137
substitution, Tay-Sachs disease, 4:101
- Glycoaldehyde phosphate, and ribose synthesis, 2:23**
- Glycogen, function, 3:41**
- Glycogen storage diseases, symptoms and treatment, 3:40, 3:44**
- Glycolipids, defined, 4:4**
- Glycolysis**
ATP production, 3:52
defined, 3:43
metabolic disorders, 3:43–44
- Glycoproteins**
blood group substances, 1:83
defined, 1:82, 3:178, 4:165
HIV and, 2:151, 2:152
in prions, 3:188
structure and function, 1:113
- Glycosidic bonds, 1:216, 1:216**
- Glycosylation**
congenital disorders, 3:40
of proteins, 3:178–179, 3:180, 3:205
- Glypican 3 proteoglycan, 2:132**
- GM foods. *See* Genetically modified foods**
- GM2 gangliosides, and Tay-Sachs disease, 4:98–100, 4:99, 4:102**
- GM-CSF (granulocyte monocyte colony stimulating factors), as gene therapy tools, 2:79**
- GNG3 (seipin) gene, and diabetes, 1:211**
- Goddard, Henry Herbert, 2:18, 2:19**
- Gold, hyperaccumulators of, 1:61**
- Goldschmidt, Richard, 3:102–103**
- Golgi apparatus, 1:105, 1:113**
defined, 4:100
- Gonadotropins**
assisted reproductive technologies, 4:19, 4:21
and endocrine disorders, 2:129
- Gonads**
defined, 1:21, 2:185, 4:78
development, AIS, 1:23–26
development, normal human, 1:21, 1:22, 1:23
hypogonadism, 1:79
See also Ovaries; Sex determination; Testes
- Gonorrhea, antibiotic resistance, 1:26, 1:27, 1:28**
- Goodfellow, Peter, 4:80**
- Gout, and uric acid, 3:44**
- Gp41 protein, HIV, 2:152, 2:152**

- Gp120 protein, HIV, 2:152, 2:152
- GPCR (G protein coupled receptor) pathway, signal transduction role, 4:88–89, 4:88, 4:90
- Gram negative bacteria
defined, 1:27
LPS layer, 2:11–12
as pathogens, 2:12
- Gram positive bacteria
Clostridium perfringens, 2:12
defined, 2:13
LPS layer, 2:11–12
as pathogens, 2:13
- Grant, Madison, 2:19
- Granulocyte monocyte colony stimulating factors (GM-CSF), as gene therapy tools, 2:79
- Granulocytes, cancerous tumors, 1:94
- Great Salt Lake, Archaea adaptations, 1:37
- Green cone pigment (GCP) genes, 1:171
- Green fluorescent protein (GFP), 2:73, 3:16, 3:17–18
- Greenhouse gases, methane, 1:37
- Greenland, ice sheet analysis, 3:168
- Greider, Carol, 4:105
- Griffith, Frederick, transformation studies, 4:122, 4:123–124
- Griffith, J.S., 3:188
- GroEl chaperones, 1:117
- Growth disorders, 2:129–132
Beckwith-Wiedemann syndrome, 2:132, 2:185
collagen and, 2:131–132, 2:201
diastrophic dysplasia, 2:38
dwarfism, 1:75–76, 2:82, 3:186, 4:1
excessive growth, 2:132
gene therapy, ethical issues, 2:82
skeletal dysplasias, 2:130–132
- Growth factor receptors
mutations, 2:130, 3:129–130
RTK pathway, 4:87–88
- Growth factors
as extranuclear hormones, 2:161
gene expression role, 2:66, 4:113–114
IGF2, 2:132, 2:185, 2:211, 3:48
oncogene impact on, 3:128, 3:129–130
signal transduction role, 4:85
- Growth hormone (GH)
and endocrine disorders, 2:129–130
in genetically modified salmon, 4:130
from recombinant DNA, 2:125
- Growth regulators, imprinting and, 2:186
- GTG banding of chromosomes, 1:127
- GTP (guanosine triphosphate), structure, 3:115, 3:117
- Guanine
concentration, for *in situ* hybridization, 2:187
in CpG dinucleotides, 3:46–49, 3:47
depurination, 1:240
in DNA alphabet, 2:83, 4:106
and DNA structure, 1:215–220, 1:250–251, 2:50, 2:51, 3:94
evolution of, 2:22–25
mutagenic base analogs, 3:87
origin of term, 1:249
pronated, 1:218
reactions with aflatoxins, 1:245
in RNA alphabet, 4:46–47, 4:106
structure, 1:216, 3:115, 3:116, 3:118, 3:118, 3:119, 4:48, 4:49
See also Base pairs
- Guanine diphosphate and triphosphate (GDP and GTP)
signal transduction role, 4:88, 4:89–90
structure, 3:115, 3:117
- Guidelines for Research Involving Recombinant DNA Molecules*, 3:152
- Guillardia*, nucleomorphs, 3:136
- Gurdon, John, 1:158, 1:161–162
- Guthrie test, for phenylketonuria, 3:176
- Gynandromorphs, 4:82
- Gynecomastia, defined, 1:79
- Gypsies, killed by Nazis, 2:20
- H**
- H antigen, 2:8
- H substance, 1:83
- HAART (highly active antiretroviral therapy), 2:155
- Habitat loss, implications for wild populations, 1:186–187, 1:188, 1:190
- Habitat use, as species isolation mechanism, 4:93
- Haemophilus influenzae*
competence development, 4:121
genome characteristics, 1:142, 1:143, 2:121, 2:172
- Hagedoorn, A.L. and A.C., 3:102
- Haldane, John Burdon Sanderson, 3:13
- Haldane map functions, 3:13
- Halobacterium*
adaptations to salty environments, 1:37
genome characteristics, 2:121
- Hamer, Dean, 4:84
- Hanta fever, causal organism, 4:165
- Haplodiploidy, in insects, 2:113
- Haploid
defined, 3:163, 4:176
gametes as, 3:24–29, 3:60, 3:163
- Haploid complements, C values, 2:113, 2:114
- Haploin-sufficiency, 2:201
- Haplotype analysis, founder populations, 2:38
- Haplotypes
defined, 2:38, 3:140
represented in pedigrees, 3:140
- Hardy, Godfrey, 2:133
- Hardy-Weinberg equilibrium, 2:133–136
allele frequency calculations, 2:134–135, 3:172–173
assumptions, 2:133–134, 3:173–174
basic concepts, 2:133
Chi-square test application, 3:195
departures from, 2:135–136
dominant alleles frequencies, 2:201
as eugenics rebuttal, 2:20
- Harley, Calvin, 4:105
- HATs (histone acetyltransferases), 1:137–138
- Hayflick, Leonard, 4:105
- Hayflick Limit, 4:104–106
- HCG (human chorionic gonadotropin), 3:185, 4:21, 4:24
- Health care and biotechnology, ethics, 1:68–69
- Health Insurance Portability and Accountability Act (HIPAA), 2:94, 3:191–192
- Hearing loss
aging and, 1:7
autosomal dominant disorders, 2:201
congenital, 1:76
diabetes and, 1:211
Martha's Vineyard population, 2:37
- Hearing, role of alternative splicing, 1:12, 1:13–14
- Heart disease. *See* Cardiovascular disease
- Heat-shock protein chaperone systems (Hsp70 and Hsp60), 1:116, 1:117, 1:118

- Heavy chain polypeptides, 2:179, 2:179
- Hedgehog proteins, 3:181
- Height, as quantitative trait, 4:1, 4:2
- HeLa, human cell line, 4:104–105
- Helicase-primase enzymes, to treat herpes, 4:13
- Helicobacter pylori*, genome characteristics, 1:142
- Helix-loop-helix motif, 3:201
- Helix-turn-helix DNA-binding motif, 4:115, 4:116
- Hematopoiesis, defined, 1:103, 2:163
- Hematopoietic cells, replication, 1:103
- Hematopoietic stem cells, 1:103, 2:76, 4:106
- Heme proteins, 2:136, 2:139, 3:202
- Hemings, Sally, 3:65
- Hemochromatosis
- clinical features, 2:202
 - and diabetes, 1:212
 - inheritance patterns, 2:203
- Hemoglobin
- amino acid alterations, 2:137, 2:139, 2:148, 3:200
 - genes coding for, 2:67–68, 2:68, 2:69
 - Hba, 2:136–138, 2:148
 - Hbc, 2:139
 - Hbf (fetal), 2:136, 2:138, 2:140
 - mutations, thalassemias, 2:139–140, 2:148
 - orthologs, 2:158
 - pseudogenes, 3:212, 3:212
 - S (Hbs), 2:137–138, 2:137, 2:148
 - structure and function, 2:136, 2:139, 3:199, 3:202
 - See also* Globin proteins; Sickle-cell disease
- Hemoglobin electrophoresis, 3:176
- Hemoglobinopathies, 2:136–141
- balanced polymorphism, 1:46
 - blood cancers, 1:93, 1:94
 - coagulation factor IX, 1:163–164
 - cytogenetics analysis, 1:119
 - genetic testing for, 3:177
 - genetics of, 2:137
 - hemoglobin C disease, 2:139
 - hemoglobin SC disease, 2:139
 - heterozygote advantage, 2:135, 2:141, 2:148
 - point mutations and, 2:27
 - thalassemias, 2:27, 2:139–141, 2:148, 4:11
 - transgenic microorganisms to combat, 1:63
 - See also* Hemophilia; Sickle-cell disease
- Hemolysis, defined, 1:85
- Hemolytic anemia, defined, 1:85
- Hemophilia, 2:141–146
- Alu sequences and, 4:146
 - clotting factor replacement, 2:145
 - DDAVP to treat, 2:145
 - detection techniques, 2:143
 - gene therapy, 2:73, 2:78, 2:80, 2:145
 - genetic components, 2:142–143
 - genetic counseling, 2:143
 - genetically engineered clotting factors, 4:6
 - inheritance patterns, 2:142, 2:203, 3:78, 4:174
 - prenatal genetic testing, 3:186
 - prophylaxis treatment, 2:145
 - Queen Victoria as carrier, 2:142, 2:144, 4:174
 - severity differences, 2:143–144
 - statistics, 2:142
 - symptoms, 2:141–142, 2:202
 - transposons and, 2:31, 2:143, 3:97
- Hemorrhagic fever, causal organism, 4:165
- Hepadnaviruses, 4:168
- Hepatitis delta virusoid, 4:164
- Hepatitis viruses
- characteristics, 4:165
 - clinical testing tools, 1:64
 - diagnostic tests, 4:6
 - hepatitis delta virusoid and, 4:164
 - reverse transcriptase, 4:41
 - vaccines to combat, 1:72, 1:255, 4:6, 4:129, 4:171
- Hepatocellular carcinoma, causal organism, 4:165
- Hepatocyte nuclear factor (HNF) genes, and diabetes, 1:211
- Herbicide-resistant crops, 1:57, 1:58, 2:106, 4:131–132, 4:134
- Herbivores, defined, 1:57
- Hereditary Genius* (F. Galton), 2:17
- Hereditary non polyposis cancer (HNPPC), 1:167, 1:169
- Heritability
- defined, 2:26, 3:144
 - estimates, defined, 4:83
 - of quantitative traits, 4:3
 - See also* Inheritance patterns; *specific diseases and traits*
- Hermaphroditic, roundworms as, 4:62
- Herpes simplex virus (HSV)
- as gene therapy vector, 2:76
 - helicase-primase enzymes to treat, 4:13
 - PML body mutations, 3:124
 - in selectable markers, 3:17
 - TK gene, 4:125
- Herpes simplex virus thymidine kinase gene (*HSV-TK*), 2:79
- Herpes zoster virus, characteristics, 4:165
- Herpesviruses
- infection outcomes, 4:169
 - replication, 4:166, 4:169
 - vaccines to combat, 4:171
- Hershey, Alfred, DNA as transforming factor, 3:104, 3:105, 4:120
- Hertwig, Oskar, 1:249
- Hessian fly, wheat resistant to, 1:58
- Heston, Leonard, 4:83
- Heterochromatin
- banding techniques for, 1:128, 1:135
 - constitutive *vs.* facultative, 1:134–135
 - defined, 1:114, 4:8
 - pericentric, 1:114
 - repetitive DNA sequences, 1:235
 - satellite sequences, 4:7
 - structure and function, 3:121–122
- Heterochromatin-euchromatin conversion, 2:63, 2:65
- Heterodimers, defined, 2:163
- Heterokaryosis, defined, 2:115
- Heteroplasmic organelles
- defined, 2:197
 - mitochondrial disease role, 3:54
- Heterosexuality. *See* Sexual orientation
- Heterosis. *See* Heterozygote advantage
- Heterotopic pregnancies, 4:22
- Heterotrophs, described, 2:13
- Heterozygote advantage, 2:146–149
- agricultural significance, 2:146
 - dominance hypothesis, 2:147
 - glucose-6-phosphate dehydrogenase deficiency, 2:148
 - Hardy-Weinberg predictions, 2:135
 - hemoglobinopathies, 2:135, 2:141, 2:148
 - in humans, 2:147–148
 - in maize, 3:10
 - overdominance hypothesis, 2:147
 - sickle-cell disease and, 1:46, 2:27, 2:135, 2:141, 2:148, 4:68
 - Tay-Sachs disease, 4:101
- Heterozygotes, defined, 1:147, 2:40
- Heterozygous
- defined, 1:35, 2:5, 3:9
 - in Mendelian genetics, 3:33–34

- HEXA* and *HEXB* genes, Tay-Sachs disease, 4:98, 4:99, 4:99, 4:100–101
- Highly active antiretroviral therapy (HAART), 2:155
- High-performance liquid chromatography (HPLC), 2:165–167
advantages, 2:165
applications, 2:165–167, 2:167, 3:197–198
components, 2:165
denaturing (DHPLC), 2:165–167
procedure, 2:165, 2:166
- High-throughput screening (HTS), 2:149–150
combinatorial chemistry role, 1:175, 1:176
procedure, 2:149–150
proteomics role, 3:205–206
uses, 2:150
- HIPAA (Health Insurance Portability and Accountability Act), 2:94, 3:191–192
- Hiroshima (Japan), radiation-caused mutations, 2:171, 2:173
- Hirschprung disease, inbreeding and, 2:191
- His*- bacteria, 1:19–20
- Hispanics
Alzheimer's disease, 2:56
cystic fibrosis, 1:202
diabetes, 1:210
- Histidine (His)
genetic code, 2:85, 4:137
bis- bacteria, 1:19–20
in transcription factors, 4:115
- Histocompatibility, defined, 2:181
- Histocompatibility leukocyte antigen (HLA) genes, and diabetes, 1:209–210
- Histograms, 4:95
- Histological, defined, 4:182
- Histone acetyltransferases (HATs), 1:137–138
- Histone deacetylases, 1:138, 2:164
- Histone proteins
acetylation of, 4:111
defined, 2:163, 3:77, 4:111
evolution of, 3:94, 3:96
function, 3:199, 4:111
gene clusters, 1:132
hormonal regulation of, 2:163–164
lacking in bacteria, 1:141, 2:15
methylation of, 3:77
MHC, immune function, 2:181, 2:181, 2:183, 4:139–140, 4:141
movement within nucleus, 3:123
in nucleosome complexes, 1:126, 1:133–134, 4:111, 4:114
octamers, 1:133
phosphorylation of, 4:111
tolerance of mutations, 2:158
- Hitachi Corporation, automated sequencers, 1:45
- Hitler, Adolf, eugenics and, 2:19–20
- HIV/AIDS, 2:150–156
acquired immune deficiency syndrome, 2:151, 2:155
antisense nucleotide tools, 1:30, 1:31
congenital, statistics, 1:75
diagnostic tests, 1:64, 4:6
DNA vaccines, 1:255
drug therapies, 2:150–151, 2:154, 2:155–156, 4:41
entering cells, 2:151–152, 2:152, 2:154
gene therapies, 2:74, 2:76, 2:80, 4:46
genetic components, 1:213
HIV structure, 2:151–152, 2:152, 4:165
immune-system impairment mechanisms, 2:153, 2:155
life cycle, 2:154
mutations, 2:155, 2:157
provirus dormancy, 2:152
retroviral role, 2:151, 4:34, 4:35, 4:38, 4:40–41, 4:52
reverse transcriptase and, 2:151, 2:152–153, 2:154, 2:155, 4:40–41
statistics, 2:150
Tat and Rev proteins, 2:153
transcription, 2:153, 2:154, 2:155–156
translation (reproduction), 2:153, 2:154
viral integrase, 2:152–153, 2:155
- HLA (human leukocyte antigen). *See* Major histocompatibility complex
- HNF* (hepatocyte nuclear factor) genes, and diabetes, 1:211
- HNPCC (hereditary non polyposis cancer), 1:167, 1:169
- Holmes, Justice Oliver Wendell, on compulsory sterilization, 2:18
- Holoprosencephaly, 2:130
- Homeobox sequence, 1:208
- Homeodomains, 1:208
- Homeostasis, defined, 2:160
- Homeotic genes and proteins
evolution of, 2:68
in fruit flies, 1:207–208, 2:45, 2:65–66, 2:68
Hox genes, 1:208
transcription factors and, 4:113–114, 4:115
- Homocystinuria, symptoms and treatment, 3:40
- Homodimers
defined, 2:162
transcription role, 2:159
- Homogentistic acid, 3:43
- Homologous chromosomes (homologues)
defined, 1:45, 2:52, 3:5, 4:121
of degenerate PCR primers, 3:158–159
evolutionary relationships, 2:156–158
of model organisms, 2:122, 3:61
point mutations and, 2:27–28
See also Crossing over (recombination); Meiosis; Mitosis; Nondisjunction
- Homologous recombination
as chromosomal repair mechanism, 2:71
gene targeting, 2:71–72, 2:72, 2:73, 4:125–126
to modify ES cells, 2:4, 2:73
- Homologous structures, evolutionary relationships, 2:156–158
- Homology, 2:156–158
computer analysis of, 2:157
defined, 2:137
to study mutation mechanisms, 2:157–158
- Homoplasmic organelles, 2:197
- Homosexuality. *See* Sexual orientation
- Homozygosity mapping, of inbred populations, 2:190–191
- Homozygous
defined, 1:35, 2:5, 3:171, 4:68
fixed alleles, 3:171
in Mendelian genetics, 1:147, 3:33–34
- Hood, Leroy, 1:44, 2:172
- Hoogsteen base pairs, 1:218, 1:218
- Horizontal evolution, 1:27
- Hormonal regulation, 2:158–165
AIS, 1:21–26
endocrine disorders, 2:129–130
hormone concentration role, 2:164
role of, 2:158–160
of transcription, 2:159, 2:162, 2:163–164
- Hormone receptors
function, 2:161
movement within nucleus, 3:123
nuclear hormone receptors, 2:161–164

- pleiotropic effects on, 3:154
transcription factors and, 4:113–114, 4:115
- Hormone therapy
estrogen replacement, 1:90
to treat Turner's syndrome, 1:79
- Hormones
autocrine, 2:160
concentration, 2:164
defined, 1:110, 3:206, 4:60
distances from source, 2:160
extracellular, 2:161
follicle-stimulating, 4:19, 4:21–22
gene expression role, 2:66–67, 4:113–114
growth, 1:63, 2:129–130
paracrine, 2:160
processing, 3:204
reproductive, and breast cancer, 1:89–91
sex-specific, 4:78–79, 4:82
signal transduction role, 1:110, 2:164
synthesis and degradation, 2:164
- Horner's syndrome, eye color changes, 2:33
- Howard Hughes Medical Institute, Human Genome Project role, 2:173
- Hox genes, 1:208, 3:107
- HPLC. *See* High-performance liquid chromatography
- HPRT (hypoxanthine phosphoribosyltransferase), 3:44
- Hprt* gene, and somatic mutations, 3:93
- HPV (human papillomavirus)
and cancer, 1:92, 4:155, 4:169
characteristics, 4:165
vaccines to combat, 4:171
- Hsp70 and Hsp60 (heat-shock protein chaperone systems), 1:116, 1:117, 1:118
- HSV. *See* Herpes simplex virus
- HSV-TK* (herpes simplex virus thymidine kinase gene), 2:79
- HTERT (human telomerase reverse transcriptase), 4:106
- HTLV (human T-lymphotropic virus), 4:34, 4:36, 4:38, 4:169
- HTS. *See* High-throughput screening
- HTS (high-throughput screening), proteomics role, 3:205–206
- Hu, Nan, 4:84
- Hudson River, PCB cleanup, 1:59
- Hughes, Austin, 2:69
- HUGO (Human Genome Organization), 3:23
- Human Cancer Genome Anatomy Project, 2:175
- Human chorionic gonadotropin (hCG), 3:185, 4:21, 4:24
- Human cloning, ethical issues, 1:69, 1:158–160, 1:161, 2:3, 2:6, 3:3
- Human disease genes, identification of, 2:167–170
candidate gene approach, 2:170
disease symptoms, 2:168
family studies, 2:169
genomic screening, 2:169–170
genomics industry role, 2:124–125
- Human Genome Project contributions, 2:102, 2:118, 2:119, 2:171
mapping studies, 3:11
McKusick's contributions, 3:23
Mendelian inheritance ratios, 3:36
mouse models as tools, 3:62
normal function of gene, 2:170
pedigrees as tools, 3:7, 3:36, 3:138–142
pharmacogenomics research, 3:146–147
polymorphisms as tools, 3:162
population screening, 1:202–203, 2:6–7, 2:99–100, 2:143, 3:175–178
recombinant DNA tools, 4:6
restriction enzyme tools, 4:33–34
SNP tools, 2:118–119, 2:166–167, 3:146–147
transgenic animal tools, 4:124, 4:127
See also Diseases, genetics of; Linkage and recombination (linkage analysis); Metabolic diseases; Mitochondrial diseases; *specific diseases and disorders*
- Human Gene Mapping Workshop, 3:23
- Human genetic diversity. *See* Individual genetic variation
- Human Genetics Laboratory, 4:80
- Human genome
alternative splicing, 1:11–12, 3:205, 4:59
archaic, DNA analysis, 3:68–69
chromosomal translocations, 3:96–97
chromosomes, number, 2:113, 2:174, 2:199, 3:64, 3:76, 3:106, 3:108–109, 3:108, 3:182, 4:13
as diploid, 2:27–28, 3:60, 3:97
distinguished from mice, 3:96, 3:97, 4:60
distinguished from other organisms, 2:122
distinguished from primates, 3:63, 3:64–65
DNA methyltransferases, 3:46, 3:49
endonuclease enzymes, 3:114
gene families, 2:29
gene loci, number, 3:107
gene sizes, 2:52
genes, number, 2:115, 2:122, 2:128, 2:174, 3:5, 3:24, 3:106, 3:205, 4:7
genetic microchips, 2:119–120, 2:119
genome size, 2:121
genotype frequencies, 3:171–174
haploid complement, 2:114
imprinted genes, 2:183
loci, number, 2:7
mapping, 3:7–8, 3:15
mitochondria, 2:116
mtDNA, 3:53–54
mutation rates, 3:99, 3:177
naming genes, 3:107–108
noncoding genes, 2:115, 2:174, 3:65, 3:83, 3:99, 4:128
nucleotide pairs, number, 3:155, 4:7
overlapping genes, 3:135–136
in pharmacogenomics studies, 3:144–147
polymorphisms, number, 2:122
repetitive sequences, 4:7–12, 4:8–9, 4:149
retropseudogenes, number, 4:10
reverse transcriptase, 4:41
rRNA, number, 3:123
SNPs, 2:191, 3:162
telomeric ends, 4:104
transposable genetic elements, 2:30, 2:67–68, 2:115, 3:64, 4:9–10
triploids, 3:164–165
See also Alu sequences; Autosomes; Sex chromosomes
- Human Genome Diversity Project, 2:175
- Human Genome Organization (HUGO), 3:23
- Human Genome Project, 2:171–178
accomplishments, 2:120, 2:121–122, 2:174–175
artificial chromosome tools, 1:145–146
BAC library tools, 1:222
behavior genetics, 1:48
bioinformatics, 1:52, 1:54

- Human Genome Project (continued)
 clinical geneticist role, 1:149–151
 code sample, 2:113
 computational biologist role, 1:181
 data acquisition, 1:54
 disease genetics contributions, 1:214–215, 2:102, 2:118, 2:119, 4:75
 drug discovery applications, 1:54
E. coli model, 1:142
 ethical issues, 2:174, 2:177
 as gene cloning tool, 1:157
 gene discovery role, 2:60
 gene therapy contributions, 2:83, 2:92
 genetic testing contributions, 2:99–100, 3:176, 3:187, 3:216
 genomics industry role, 2:123–125
 history, 1:73–74
 individual variation questions, 2:175
 McKusick's contributions, 3:23
 mtDNA compared to nuclear DNA, 3:57
 origin, 2:171–173
 patenting the genome, 2:174, 2:175–176
 pharmacogenomics applications, 3:146–147
 protein sequencing contributions, 3:198
 proteomics and, 2:177
 public and private sector competition, 2:173–174
 racial and geographic differences, 2:175
 recombinant DNA tools, 4:6
 Watson's contributions, 2:173–174, 2:175, 2:176, 4:173
 Working Group on Ethical, Legal, and Social Implications of, 2:94
- Human Genome Sciences, gene sequence data, 2:123
- Human growth hormones
 antibodies against, 1:255
 genetically engineered, 1:63, 1:153, 4:6
- Human Immunodeficiency Virus. *See* HIV/AIDS
- Human leukocyte antigen (HLA). *See* Major histocompatibility complex
- Human papillomavirus (HPV)
 and cancer, 1:92, 4:155, 4:169
 characteristics, 4:165
 vaccines to combat, 4:171
- Human PSD database, 3:209
- Human telomerase reverse transcriptase (hTERT), 4:106
- Human T-lymphotropic virus (HTLV), 4:34, 4:36, 4:38, 4:169
- Humans (*Homo* sp.)
 archaic, 3:66–67
 divergence from chimpanzees, 3:167–168, 3:168
 geographic origin, 3:64, 3:66–67, 3:66, 3:167–170, 3:174, 4:179
H. erectus distinguished from Neandertals, 3:66–67, 3:66, 3:69, 3:167
 replacing Neandertals, 3:170
See also Anthropology, molecular
- Hunkapiller, Michael, 1:44
- Hunter syndrome (iduronate sulfatase deficiency), symptoms and treatment, 3:41, 3:45
- Huntingtin* gene, 1:179, 4:151, 4:152
- Huntington's disease
 apoptosis and, 1:33
 Chi-square test to predict, 3:194–195
 clinical features, 2:202, 4:151
 genetic counseling, 2:103–104
 genetic testing, 2:100, 2:101, 2:103–104, 2:168
 inheritance patterns, 2:104, 2:127, 2:201
 mutation rates, 3:100
 pharmacogenomics to predict, 3:146
 as polyglutamine disorder, 4:151
 protein conformation role, 3:190
 as simple trait, 1:177, 1:179
 as toxic-gain-of-function mutation, 2:201
 as triplet repeat disorder, 2:42, 3:96, 4:4, 4:11
- Hurler syndrome (α -iduronidase deficiency), symptoms and treatment, 3:41, 3:45
- Hutchinson-Gilford syndrome, 1:2
 diagnosis, 1:1, 1:3
 telomere shortening, 1:3
- Hutterite colonies, asthma, 2:37, 2:37, 2:38–39
- Huxley, Aldous, 4:26
- Hybrid superiority (hybrid vigor). *See* Heterozygote advantage
- Hybridization (molecular). *See In situ* hybridization
- Hybridization (plant and animal breeding)
Arabidopsis thaliana, 1:33–35
 as artificial selection, 4:67, 4:68–69
- classical hybrid genetics, 1:146–149
 distinguished from transgenic organisms, 2:106
 extranuclear inheritance role, 2:194
 heterozygote advantage, 2:146–147
 inbreeding, 2:189–190
 of maize, 3:8–10
 plant genetic engineer role, 3:149–150
 polyploidy, 2:114
 quantitative trait role, 4:1
See also Mendelian genetics
- Hybrids/hybridize, defined, 1:11, 4:68
- Hydrogen bonds
 defined, 1:215, 2:24, 3:119, 4:12
 and DNA structure, 1:215–217, 1:217, 1:251, 2:24, 2:51, 2:85, 2:186, 3:116, 3:119, 4:12, 4:172
 from DNA to gene regulatory proteins, 2:64
 and protein secondary structure, 3:200–201
 and RNA structure, 2:24, 2:186, 4:47
- Hydrogen cyanide, and adenine synthesis, 2:23
- Hydrolysis
 defined, 1:117
 errors, DNA damage, 1:240, 1:240
- Hydrolysis reactions, of nucleases, 3:113, 3:114
- Hydrophilic, defined, 1:109, 3:200
- Hydrophobic, defined, 1:109, 1:217, 3:200
- Hydroxylation of proteins, post-translational, 3:203
- Hydroxyurea, to treat sickle-cell disease, 2:138
- Hyper- and hypoacetylation of histones, 1:137–138
- Hyperaccumulator plants, 1:60, 1:62
- Hyperactivity. *See* Attention deficit hyperactivity disorder
- Hyperammonemia, 3:45
- Hypercholesterolemia, 3:40, 3:43, 4:4–5
- Hyperhomocysteinemia, symptoms and treatment, 3:40
- Hyperlipidemia, symptoms and treatment, 3:40, 3:43
- Hyperplastic cells, defined, 1:95
- Hypertension
 and cardiovascular disease, 1:101
 defined, 1:101
 populations studies, 2:124

- Hypertrophic muscles, 3:84
- Hypervariable regions, antibodies, 2:179, 2:179
- Hyphae, fungal, 2:115
- Hypochondroplasia, characteristics, 2:130
- Hypogonadism, defined, 1:79
- Hypopituitarism, 2:129–130
- Hypoplasia, characteristics, 2:130
- Hypothalamus
defined, 4:84
sexual orientation role, 4:84
- Hypotheses
alternate, 4:96–97
defined, 3:7, 3:18
null, 4:96–97
testing, 4:96–97
- Hypothyroidism, population screening, 3:176–177
- Hypoxanthine phosphoribosyltransferase (HPRT), 3:44
- I**
- ICD10 clinical criteria, 1:4
- Ice ages, population bottlenecks, 3:168
- Iceland
founder effect, 2:37
populations studies, 2:124
- ICF (Immunodeficiency, Centromere Instability, and Facial Anomalies) syndrome, 3:49
- Icosahedral symmetry, of viruses, 4:165
- Idiopathic torsion dystonia, founder effect, 2:38
- Iduronate sulfatase deficiency (Hunter syndrome), 3:41, 3:45
- α -iduronidase deficiency (Hurler syndrome), 3:41, 3:45
- IGF2. *See* Insulin-like growth factor II
- Immigration and migration
of early humans, 3:62, 3:64, 3:66–67, 3:70, 3:168–170
eugenics and, 2:16, 2:19
gene flow and, 2:70–71
in Hardy-Weinberg model, 2:133
limited, founder effect, 2:36–39, 3:168–169, 3:169
patterns, extranuclear genes to study, 2:198
- Immune system disorders (immunodeficiency)
acquired, 4:74
primary, 4:74–75
See also HIV/AIDS; Severe combined immune deficiency (SCID)
- Immune system genetics, 2:178–183
of cheetah populations, 1:189
combinatorial chemistry tools, 1:174
diabetes type 1, 1:209–210
diversity, 2:128, 2:179
DNA methylation role, 3:47–48, 3:49
Down syndrome, 1:3
MHC proteins, 1:8, 2:181, 2:181, 2:183, 3:82, 4:139–140, 4:141
random recombination of gene segments, 2:180–181
restriction enzyme role, 4:31
Rh blood incompatibility, 1:85–86
RNA interference role, 4:52–54
SMCY genes, and transplant rejection, 4:177
somatic recombination, 2:180, 2:182
- Immune system response
antiviral vaccines, 4:170–171
lipid vesicle vectors, 2:76–77
plasmid vectors, 2:78
recombinant adenovirus vectors, 2:75, 2:77, 2:78
retroviral infections, 4:38
retroviruses, 2:76
transgenes, 2:77
tumor tissue, 2:79
- Immune systems
autocrine hormones, 2:160
B cells, 2:178–181
CD4T cells, 2:152, 2:153, 2:154, 2:155, 4:36
E. coli to study, 2:11
embryonic stem cells and, 2:6
function, 2:178, 4:74
immunoglobulin proteins, 2:29
macrophages, 2:178
natural killer cells, 4:75
nonspecific *vs.* specific immunity, 2:178
rodent models to study, 4:60
vaccination principles, 1:253–254
See also Antibodies; Antigens; T cells
- Immunodeficiency, Centromere Instability, and Facial Anomalies (ICF) syndrome, 3:49
- Immunogenicity, defined, 2:78, 4:140
- Immunoglobulin proteins, defined, 2:29
- Immunosuppressive, defined, 4:38
- Immunosuppressive drugs, to prevent organ rejection, 4:141
- Importins, 3:126
- Imprinting, 2:183–186, 3:48
distinguished from dominant/recessive alleles, 2:183–184
distinguished from sex-linked genes, 2:183
DNA methylation role, 3:48
evolutionary advantages, 2:185–188
gene expression and, 2:183–184, 2:201, 3:48
IGF2 gene example, 2:185, 3:48
inheritance patterns, 2:205
parental genomic, 2:183
timing and mechanism of, 2:184, 2:184, 2:185
uniparental disomy, 2:185
- In situ* hybridization (molecular), 2:186–189
in blotting procedures, 1:87–89
defined, 1:87
detection tools, 2:187–188
in DNA microarrays, 1:225–226
oncogene detection role, 3:129
PCR as tool, 3:158
as prenatal diagnostic tool, 3:182–183
preserving cells and chromosomes, 2:188
procedure, 1:136–137, 2:186–187
specificity control, 2:188–189
tissue and cell applications, 2:187
in transposon sequencing, 2:43
See also Fluorescence *in situ* hybridization (FISH)
- In vitro*, defined, 2:3, 3:18, 4:20
- In vitro* evolution, 2:25
- In vitro* fertilization
age concerns, 4:21, 4:23, 4:25
antidiscrimination laws, 4:29
cystic fibrosis and, 1:201
embryonic stem cells, 2:3–4
ethical issues, 1:69, 4:28–29
microassisted, 4:22
preimplantation genetic diagnosis, 2:98, 3:182, 3:184, 3:185–187
procedures, 2:36, 4:21–23
risks, 4:22
screening process, 4:20–21
success rate, 4:20, 4:23
- In vitro* hybridization, 2:186–187
- In vivo*, defined, 1:116
- Inbreeding, 2:189–191
benefits, 2:190
defined, 1:188, 2:146

- Inbreeding (continued)
 and disease risk, 2:189–190, 2:203, 3:174
 founder effect, 2:36–39, 3:174
 laws that prevent, 2:189
vs. outcrossing, fitness, 2:146
 polymorphisms to study, 3:163
 populations studies, 2:190–191
- Inbreeding coefficients, defined, 2:190
- Inbreeding depression
 agricultural significance, 2:146
 in conservation biology, 1:188–189
 explanations for, 2:146–147
 polyploidy and, 2:69
- Incomplete dominance, 2:200, 3:36
- Incomplete penetrance, 2:201, 2:205
- Incubating, defined, 1:20
- Incyte, gene sequence data, 2:123
- Indels (insertion or deletion polymorphisms), 3:160
- Indians (American), evolutionary origin, 3:66
- Individual genetic variation, 2:191–192
 allele scoring strategies, 2:192
 alleles and, 2:191–192
 alternative splicing and, 1:11–12
 among early humans, 3:167–168, 3:168
 and DNA profiling, 3:192–193
 examples, 2:191, 2:192, 2:192, 2:202
 genetic microchips, 2:119–120, 2:119
 Human Genome Project concerns, 2:175
 pharmacogenomics applications, 3:144–147
 selection and, 4:69
vs. speciation, 4:91–92
- Industrial toxins
 and birth defects, 1:77
 and cancer, 1:92
 and chromosomal aberrations, 1:119, 1:123
 and diabetes, 1:210
 and DNA damage, 1:239, 1:242, 3:89
 role in disease, 1:214
See also Mutagens
- Infertility
 causes of, 4:19–20
 chromosomal translocations, 3:97
 drugs to treat, 4:19
 in male partner, 4:20, 4:21, 4:177
 statistics, 4:19
 surgical interventions, 4:19
See also Reproductive technology
- Inflammatory bowel disease, and colon cancer, 1:170
- Influenza virus
 characteristics, 4:165
 DNA vaccines, 1:255
 pathogenicity of, 2:128
- Informants, in pedigree drawing, 3:140–141
- Information systems managers, 2:192–194
- Informed consent
 defined, 2:90, 3:218
 family studies, 2:168
 gene therapy clinical trials, 2:81–82
 genetic counselor role, 2:90
 genetic testing, 2:102, 3:175
- Ing1 retrotransposons, 4:145
- Ingram, Vernon, 3:95
- Inheritance, extranuclear, 2:194–199
 of intracellular symbionts, 2:198
 plasmids, 2:198, 3:151
 significance of, 2:194, 2:198, 3:57
See also Chloroplasts, genome; Mitochondrial genome
- Inheritance patterns, 2:199–207
 anticipation, 2:204–205, 4:150
Arabidopsis research, 1:34–36
 biparental, 2:197
 blending theory, 3:30, 3:34, 3:102
 blood groups, 1:83–84
 cartilage matrix protein mutations, 2:132
 chromosomal aberrations, 1:120–121, 1:125
 complex traits, 1:177–181, 1:213–214, 2:61, 2:206
 de novo mutations, 1:120
 epistatic interactions, 2:8–9, 2:207
 extranuclear, 2:195, 2:196–197, 2:204, 2:205
 gain-of-function mutations, 3:96
 gene-environment interactions, 2:56–57
 genetic counselors to explain, 2:88–90, 2:91–92
 genetic discrimination, 1:68
 growth factor receptor mutations, 2:130
 heterozygote advantage, 2:147–148
 inbreeding, 2:189–191
 incomplete penetrance, 2:201, 2:204
 intracellular symbionts, 2:198
 in linkage and recombination studies, 3:7–8
 loss-of-function genetic mutations, 3:96
 mosaicism, 2:204
 multifactorial and polygenic, 1:77–78, 2:206, 3:214
 natural selection and, 2:26
 pedigrees as tools, 3:7, 3:36, 3:138–142
 of plasmids, 2:198, 3:151
 pleiotropy, 2:206
 Punnett squares, 1:148, 3:33, 3:35, 3:193–194
 quantitative traits, 4:3
 single-gene diseases, 2:60
 single-gene mutations, 1:75
 STRs and, 1:189
 transposable genetic elements, 4:145–146, 4:145
 uniparental, 2:197
 variable expressivity, 2:201, 2:204
See also Chromosomal theory of inheritance; Classical hybrid genetics; Dominant alleles; Hardy-Weinberg equilibrium; Imprinting; Mendelian genetics; Recessive alleles; Sex-linked inheritance; X-linked disorders
- Inheritance patterns, specific diseases and traits, 1:82
 achondroplasia, 1:75–76
 addiction, 1:4–5
 ADHD, 1:40–41
 AIS, 1:21–25
 Alzheimer's disease, 1:16–18
 Beckwith-Wiedemann syndrome, 2:132
 behavior, 1:46–51
 cancer, 1:91–92, 1:97, 1:168–170
 cardiovascular disease, 1:101–103
 color vision defects, 1:171–172, 3:78
 cystic fibrosis, 1:200–203, 1:214, 3:142, 3:193–194
 diabetes, 1:82, 1:211–212
 endocrine disorders, 2:129–130
 eye color, 2:32–33
 founder effect, 2:36–37
 fragile X syndrome, 2:39–42
 hemophilia, 2:142–143, 2:144, 3:78
 Huntington's disease, 2:103–105, 2:104
 Marfan syndrome, 1:76
 metabolic diseases, 3:38–39, 3:45
 muscular dystrophies, 1:213, 2:191–192, 2:202, 2:203, 3:78, 3:83–84, 3:85, 3:86

- phenylketonuria, 2:55–56
 progeria, 1:1–3
 psychiatric disorders, 3:214, 3:215
 retinoblastomas, 4:153
 sickle-cell disease, 2:137–138, 3:142
 thalassemias, 2:140
 thyroid hormone deficiency/resistance, 2:129–130
 Y chromosomes, 4:176–177
In silico, defined, 2:212
 In utero, defined, 2:81
 In utero gene therapy, 2:81, 2:82
 In vivo, defined, 2:212, 4:43
 Initiation codons and sites. *See* Start codons (transcription)
 Initiation complexes, translation, 4:138
 Initiation factors
 transcription, 4:107
 translation, 4:138
 INK family proteins, 1:107
Inquiries into the Human Faculty and Its Development (F. Galton), 2:17
 Insect-resistant crops
 Bacillus thuringiensis, 1:58, 1:64, 3:149, 4:134
 ecological concerns, 1:68
 genetically engineered, 1:63, 2:108
 genomics role, 2:121
 statistics, 1:57
 transgenic plants, 2:106
 Insects
 ants, chromosome numbers, 2:113
 haplodiploidy, 2:113
 holocentric chromosomes, 1:114
 intracellular symbionts, 2:198
 polytene chromosomes, 1:126, 1:136–137, 1:136
 See also Fruit flies (*Drosophila*)
 Insemination
 donor, 4:24
 intrauterine, 4:24
 Insertion sequence (IS) elements, prokaryotes, 2:116, 2:117, 4:144–145
 Insertional mutagenesis, 4:38
 The Institute for Genomic Research (TIGR), 2:173
 Insulin
 as extranuclear hormone, 2:161
 function, 1:209, 3:199
 gene expression role, 2:66–67
 genetically engineered, 1:63, 1:66, 1:72, 1:153–154, 2:125, 4:6, 4:129
 post-translational control, 3:181
 RTK pathway, 4:87–88
 sequence of, 4:64
 Insulin receptors, pleiotropic effects on, 3:154
 Insulin resistance
 and diabetes, 1:210–211
 insulin receptor gene defects, 1:211
 Insulin-dependent diabetes mellitus (diabetes type 1), 1:209–210, 2:202
 Insulin-like growth factor II (IGF2) Beckwith-Wiedemann syndrome and, 2:132
 DNA methylation, 3:48
 imprinting, 2:185
 intelligence and, 2:211
 Insurance discrimination, genetic testing and, 1:18, 1:68, 2:92–94, 2:100, 2:177, 3:4, 3:146, 3:192, 3:193
 Integrase enzymes
 HIV, 2:152–153, 2:155
 of retroviruses, 4:35, 4:37
 Inteins, post-translational control, 3:179
 Intelligence, 2:207–211
 as complex trait, 2:206, 2:211, 3:81
 Down syndrome, average IQ, 1:79
 environmental factors, 2:207, 2:209–210, 2:211
 eugenics concepts, 2:18
 expanded concepts, 2:209
 Flynn effect, 2:209
 genes for, 2:5, 2:211
 IQ measurements, 2:207–208
 MENSA, 2:208
 racial and ethnic differences, 2:209
 standard tests, 2:208
 twin and adoption studies, 2:207–208, 2:209–210
 X chromosome role, 4:174
 See also Mental retardation
 Intelligence quotient tests. *See* IQ tests
 Interbreeding, 3:171, 3:173–174, 4:92–93
 Intercalating agents, 3:88
 Interchromatin compartment, 3:120, 3:121, 3:122–123, 3:122
 Interchromatin granules (speckles), 3:120, 3:125
 Intercrossing, defined, 1:147
 Interference, crossover, 1:197, 3:12–13
 Interferons
 as extranuclear hormones, 2:161
 from genetically engineered bacteria, 1:63
 Interleukin receptors, SCID and, 4:75, 4:76, 4:77
 Interleukins
 as extranuclear hormones, 2:161
 as gene therapy tools, 2:79
 from genetically engineered bacteria, 1:63
 Intermediate filaments, 3:120, 3:121
 International Congress of Eugenics, 3:81
 International Human Genome Consortium, 2:174, 2:176
 International Rice Research Institute, 2:108
 International World Conservation Union (IUCN), 1:186
 Internet, 2:211–213
 bioinformatics tools, 1:53, 3:18
 biotechnology resources, 1:64, 1:65
 gene mutation listings, 2:143
 genetic maps, 3:15
 Genome Web site, 1:142, 2:176
 Human Genome Project information, 2:174
 International Human Genome Consortium, 2:176
 Mendelian Inheritance in Man, 3:23
 Interphase, 2:112–113
 banding techniques for, 1:129
 defined, 1:134
 DNA synthesis (S) phase, 1:104–105, 1:104, 3:25–26, 3:57–59
 growth (G_1 and G_2) phases, 1:104–105, 3:57–59
 Intersex organisms, 1:25, 3:77, 3:164, 4:80, 4:82, 4:127, 4:175
 Interstitial region, 4:78
 Intervention strategies, public health role, 3:218–219
 Intestinal disorders. *See* Gastrointestinal disorders
 Intracytoplasmic sperm injection, 4:22
 Intra-strand UV crosslinks, 1:244, 1:245
 Intrauterine insemination, 4:24
 Intravenous, defined, 2:145
 Intrinsic terminators, 4:110
 Introns, 1:12, 4:58
 of antibody genes, 2:180, 2:182
 defined, 1:224, 2:30, 3:56, 4:10
 discovery of, 3:105–106